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The Laplace Transform

EXERCISES 4.1

Definition of the Laplace Transform

1. $\mathcal{L}\{f(t)\} = \int_0^1 -e^{-st} dt + \int_1^\infty e^{-st} dt = \frac{1}{s} e^{-st} \Big|_0^1 - \frac{1}{s} e^{-st} \Big|_1^\infty$
 $= \frac{1}{s} e^{-s} - \frac{1}{s} - \left(0 - \frac{1}{s} e^{-s}\right) = \frac{2}{s} e^{-s} - \frac{1}{s}, \quad s > 0$
2. $\mathcal{L}\{f(t)\} = \int_0^2 4e^{-st} dt = -\frac{4}{s} e^{-st} \Big|_0^2 = -\frac{4}{s} (e^{-2s} - 1), \quad s > 0$
3. $\mathcal{L}\{f(t)\} = \int_0^1 t e^{-st} dt + \int_1^\infty e^{-st} dt = \left(-\frac{1}{s} t e^{-st} - \frac{1}{s^2} e^{-st}\right) \Big|_0^1 - \frac{1}{s} e^{-st} \Big|_1^\infty$
 $= \left(-\frac{1}{s} e^{-s} - \frac{1}{s^2} e^{-s}\right) - \left(0 - \frac{1}{s^2}\right) - \frac{1}{s} (0 - e^{-s}) = \frac{1}{s^2} (1 - e^{-s}), \quad s > 0$
4. $\mathcal{L}\{f(t)\} = \int_0^1 (2t + 1) e^{-st} dt = \left(-\frac{2}{s} t e^{-st} - \frac{2}{s^2} e^{-st} - \frac{1}{s} e^{-st}\right) \Big|_0^1$
 $= \left(-\frac{2}{s} e^{-s} - \frac{2}{s^2} e^{-s} - \frac{1}{s} e^{-s}\right) - \left(0 - \frac{2}{s^2} - \frac{1}{s}\right) = \frac{1}{s} (1 - 3e^{-s}) + \frac{2}{s^2} (1 - e^{-s}), \quad s > 0$
5. $\mathcal{L}\{f(t)\} = \int_0^\pi (\sin t) e^{-st} dt = \left(-\frac{s}{s^2 + 1} e^{-st} \sin t - \frac{1}{s^2 + 1} e^{-st} \cos t\right) \Big|_0^\pi$
 $= \left(0 + \frac{1}{s^2 + 1} e^{-\pi s}\right) - \left(0 - \frac{1}{s^2 + 1}\right) = \frac{1}{s^2 + 1} (e^{-\pi s} + 1), \quad s > 0$
6. $\mathcal{L}\{f(t)\} = \int_{\pi/2}^\infty (\cos t) e^{-st} dt = \left(-\frac{s}{s^2 + 1} e^{-st} \cos t + \frac{1}{s^2 + 1} e^{-st} \sin t\right) \Big|_{\pi/2}^\infty$
 $= 0 - \left(0 + \frac{1}{s^2 + 1} e^{-\pi s/2}\right) = -\frac{1}{s^2 + 1} e^{-\pi s/2}, \quad s > 0$
7. $f(t) = \begin{cases} 0, & 0 < t < 1 \\ t, & t > 1 \end{cases}$
 $\mathcal{L}\{f(t)\} = \int_1^\infty t e^{-st} dt = \left(-\frac{1}{s} t e^{-st} - \frac{1}{s^2} e^{-st}\right) \Big|_1^\infty = \frac{1}{s} e^{-s} + \frac{1}{s^2} e^{-s}, \quad s > 0$
8. $f(t) = \begin{cases} 0, & 0 < t < 1 \\ 2t - 2, & t > 1 \end{cases}$
 $\mathcal{L}\{f(t)\} = 2 \int_1^\infty (t - 1) e^{-st} dt = 2 \left(-\frac{1}{s} (t - 1) e^{-st} - \frac{1}{s^2} e^{-st}\right) \Big|_1^\infty = \frac{2}{s^2} e^{-s}, \quad s > 0$

9. The function is $f(t) = \begin{cases} 1-t, & 0 < t < 1 \\ 0, & t > 1 \end{cases}$ so

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^1 (1-t)e^{-st} dt + \int_1^\infty 0e^{-st} dt = \int_0^1 (1-t)e^{-st} dt = \left(-\frac{1}{s}(1-t)e^{-st} + \frac{1}{s^2}e^{-st} \right) \Big|_0^1 \\ &= \frac{1}{s^2}e^{-s} + \frac{1}{s} - \frac{1}{s^2}, \quad s > 0 \end{aligned}$$

10. $f(t) = \begin{cases} 0, & 0 < t < a \\ c, & a < t < b \\ 0, & t > b \end{cases}; \quad \mathcal{L}\{f(t)\} = \int_a^b ce^{-st} dt = -\frac{c}{s}e^{-st} \Big|_a^b = \frac{c}{s}(e^{-sa} - e^{-sb}), \quad s > 0$

11. $\mathcal{L}\{f(t)\} = \int_0^\infty e^{t+7}e^{-st} dt = e^7 \int_0^\infty e^{(1-s)t} dt = \frac{e^7}{1-s}e^{(1-s)t} \Big|_0^\infty = 0 - \frac{e^7}{1-s} = \frac{e^7}{s-1}, \quad s > 1$

12. $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-2t-5}e^{-st} dt = e^{-5} \int_0^\infty e^{-(s+2)t} dt = -\frac{e^{-5}}{s+2}e^{-(s+2)t} \Big|_0^\infty = \frac{e^{-5}}{s+2}, \quad s > -2$

13. $\mathcal{L}\{f(t)\} = \int_0^\infty te^{4t}e^{-st} dt = \int_0^\infty te^{(4-s)t} dt = \left(\frac{1}{4-s}te^{(4-s)t} - \frac{1}{(4-s)^2}e^{(4-s)t} \right) \Big|_0^\infty$
 $= \frac{1}{(4-s)^2}, \quad s > 4$

14. $\mathcal{L}\{f(t)\} = \int_0^\infty t^2e^{-2t}e^{-st} dt = \int_0^\infty t^2e^{-(s+2)t} dt$
 $= \left(-\frac{1}{s+2}t^2e^{-(s+2)t} - \frac{2}{(s+2)^2}te^{-(s+2)t} - \frac{2}{(s+2)^3}e^{-(s+2)t} \right) \Big|_0^\infty = \frac{2}{(s+2)^3}, \quad s > -2$

15. $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-t}(\sin t)e^{-st} dt = \int_0^\infty (\sin t)e^{-(s+1)t} dt$
 $= \left(\frac{-(s+1)}{(s+1)^2+1}e^{-(s+1)t}\sin t - \frac{1}{(s+1)^2+1}e^{-(s+1)t}\cos t \right) \Big|_0^\infty$
 $= \frac{1}{(s+1)^2+1} = \frac{1}{s^2+2s+2}, \quad s > -1$

16. $\mathcal{L}\{f(t)\} = \int_0^\infty e^t(\cos t)e^{-st} dt = \int_0^\infty (\cos t)e^{(1-s)t} dt$
 $= \left(\frac{1-s}{(1-s)^2+1}e^{(1-s)t}\cos t + \frac{1}{(1-s)^2+1}e^{(1-s)t}\sin t \right) \Big|_0^\infty$
 $= -\frac{1-s}{(1-s)^2+1} = \frac{s-1}{s^2-2s+2}, \quad s > 1$

17. $\mathcal{L}\{f(t)\} = \int_0^\infty t(\cos t)e^{-st} dt$
 $= \left[\left(-\frac{st}{s^2+1} - \frac{s^2-1}{(s^2+1)^2} \right) (\cos t)e^{-st} + \left(\frac{t}{s^2+1} + \frac{2s}{(s^2+1)^2} \right) (\sin t)e^{-st} \right]_0^\infty$
 $= \frac{s^2-1}{(s^2+1)^2}, \quad s > 0$

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18. $\mathcal{L}\{f(t)\} = \int_0^\infty t(\sin t)e^{-st}dt$

$$= \left[\left(-\frac{t}{s^2+1} - \frac{2s}{(s^2+1)^2} \right) (\cos t)e^{-st} - \left(\frac{st}{s^2+1} + \frac{s^2-1}{(s^2+1)^2} \right) (\sin t)e^{-st} \right]_0^\infty$$

$$= \frac{2s}{(s^2+1)^2}, \quad s > 0$$
19. $\mathcal{L}\{2t^4\} = 2\frac{4!}{s^5}$
20. $\mathcal{L}\{t^5\} = \frac{5!}{s^6}$
21. $\mathcal{L}\{4t - 10\} = \frac{4}{s^2} - \frac{10}{s}$
22. $\mathcal{L}\{7t + 3\} = \frac{7}{s^2} + \frac{3}{s}$
23. $\mathcal{L}\{t^2 + 6t - 3\} = \frac{2}{s^3} + \frac{6}{s^2} - \frac{3}{s}$
24. $\mathcal{L}\{-4t^2 + 16t + 9\} = -4\frac{2}{s^3} + \frac{16}{s^2} + \frac{9}{s}$
25. $\mathcal{L}\{t^3 + 3t^2 + 3t + 1\} = \frac{3!}{s^4} + 3\frac{2}{s^3} + \frac{3}{s^2} + \frac{1}{s}$
26. $\mathcal{L}\{8t^3 - 12t^2 + 6t - 1\} = 8\frac{3!}{s^4} - 12\frac{2}{s^3} + \frac{6}{s^2} - \frac{1}{s}$
27. $\mathcal{L}\{1 + e^{4t}\} = \frac{1}{s} + \frac{1}{s-4}$
28. $\mathcal{L}\{t^2 - e^{-9t} + 5\} = \frac{2}{s^3} - \frac{1}{s+9} + \frac{5}{s}$
29. $\mathcal{L}\{1 + 2e^{2t} + e^{4t}\} = \frac{1}{s} + \frac{2}{s-2} + \frac{1}{s-4}$
30. $\mathcal{L}\{e^{2t} - 2 + e^{-2t}\} = \frac{1}{s-2} - \frac{2}{s} + \frac{1}{s+2}$
31. $\mathcal{L}\{4t^2 - 5\sin 3t\} = 4\frac{2}{s^3} - 5\frac{3}{s^2+9}$
32. $\mathcal{L}\{\cos 5t + \sin 2t\} = \frac{s}{s^2+25} + \frac{2}{s^2+4}$
33. $\mathcal{L}\{\sinh kt\} = \frac{1}{2}\mathcal{L}\{e^{kt} - e^{-kt}\} = \frac{1}{2}\left[\frac{1}{s-k} - \frac{1}{s+k}\right] = \frac{k}{s^2-k^2}$
34. $\mathcal{L}\{\cosh kt\} = \frac{1}{2}\mathcal{L}\{e^{kt} + e^{-kt}\} = \frac{s}{s^2-k^2}$
35. $\mathcal{L}\{e^t \sinh t\} = \mathcal{L}\left\{e^t \frac{e^t - e^{-t}}{2}\right\} = \mathcal{L}\left\{\frac{1}{2}e^{2t} - \frac{1}{2}\right\} = \frac{1}{2(s-2)} - \frac{1}{2s}$
36. $\mathcal{L}\{e^{-t} \cosh t\} = \mathcal{L}\left\{e^{-t} \frac{e^t + e^{-t}}{2}\right\} = \mathcal{L}\left\{\frac{1}{2} + \frac{1}{2}e^{-2t}\right\} = \frac{1}{2s} + \frac{1}{2(s+2)}$
37. $\mathcal{L}\{\sin 2t \cos 2t\} = \mathcal{L}\left\{\frac{1}{2}\sin 4t\right\} = \frac{2}{s^2+16}$
38. $\mathcal{L}\{\cos^2 t\} = \mathcal{L}\left\{\frac{1}{2} + \frac{1}{2}\cos 2t\right\} = \frac{1}{2s} + \frac{1}{2}\frac{s}{s^2+4}$
39. From the addition formula for the sine function, $\sin(4t+5) = \sin 4t \cos 5 + \cos 4t \sin 5$ so

$$\mathcal{L}\{\sin(4t+5)\} = (\cos 5)\mathcal{L}\{\sin 4t\} + (\sin 5)\mathcal{L}\{\cos 4t\} = (\cos 5)\frac{4}{s^2+16} + (\sin 5)\frac{s}{s^2+16} = \frac{4\cos 5 + (\sin 5)s}{s^2+16}.$$
40. From the addition formula for the cosine function,

$$\cos\left(t - \frac{\pi}{6}\right) = \cos t \cos \frac{\pi}{6} + \sin t \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2}\cos t + \frac{1}{2}\sin t$$

so

$$\begin{aligned}\mathcal{L}\left\{\cos\left(t - \frac{\pi}{6}\right)\right\} &= \frac{\sqrt{3}}{2}\mathcal{L}\{\cos t\} + \frac{1}{2}\mathcal{L}\{\sin t\} \\ &= \frac{\sqrt{3}}{2}\frac{s}{s^2+1} + \frac{1}{2}\frac{1}{s^2+1} = \frac{1}{2}\frac{\sqrt{3}s+1}{s^2+1}.\end{aligned}$$

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41. (a) Using integration by parts for $\alpha > 0$,

$$\Gamma(\alpha + 1) = \int_0^\infty t^\alpha e^{-t} dt = -t^\alpha e^{-t} \Big|_0^\infty + \alpha \int_0^\infty t^{\alpha-1} e^{-t} dt = \alpha \Gamma(\alpha).$$

- (b) Let $u = st$ so that $du = s dt$. Then

$$\mathcal{L}\{t^\alpha\} = \int_0^\infty e^{-st} t^\alpha dt = \int_0^\infty e^{-u} \left(\frac{u}{s}\right)^\alpha \frac{1}{s} du = \frac{1}{s^{\alpha+1}} \Gamma(\alpha + 1), \quad \alpha > -1.$$

42. (a) $\mathcal{L}\{t^{-1/2}\} = \frac{\Gamma(1/2)}{s^{1/2}} = \sqrt{\frac{\pi}{s}}$ (b) $\mathcal{L}\{t^{1/2}\} = \frac{\Gamma(3/2)}{s^{3/2}} = \frac{\sqrt{\pi}}{2s^{3/2}}$ (c) $\mathcal{L}\{t^{3/2}\} = \frac{\Gamma(5/2)}{s^{5/2}} = \frac{3\sqrt{\pi}}{4s^{5/2}}$

43. Let $F(t) = t^{1/3}$. Then $F(t)$ is of exponential order, but $f(t) = F'(t) = \frac{1}{3}t^{-2/3}$ is unbounded near $t = 0$ and hence is not of exponential order. Let

$$f(t) = 2te^{t^2} \cos e^{t^2} = \frac{d}{dt} \sin e^{t^2}.$$

This function is not of exponential order, but we can show that its Laplace transform exists. Using integration by parts we have

$$\begin{aligned} \mathcal{L}\{2te^{t^2} \cos e^{t^2}\} &= \int_0^\infty e^{-st} \left(\frac{d}{dt} \sin e^{t^2} \right) dt = \lim_{a \rightarrow \infty} \left[e^{-st} \sin e^{t^2} \Big|_0^a + s \int_0^a e^{-st} \sin e^{t^2} dt \right] \\ &= -\sin 1 + s \int_0^\infty e^{-st} \sin e^{t^2} dt = s \mathcal{L}\{\sin e^{t^2}\} - \sin 1. \end{aligned}$$

Since $\sin e^{t^2}$ is continuous and of exponential order, $\mathcal{L}\{\sin e^{t^2}\}$ exists, and therefore $\mathcal{L}\{2te^{t^2} \cos e^{t^2}\}$ exists.

44. The relation will be valid when s is greater than the maximum of c_1 and c_2 .
 45. Since e^t is an increasing function and $t^2 > \ln M + ct$ for $M > 0$ we have $e^{t^2} > e^{\ln M + ct} = Me^{ct}$ for t sufficiently large and for any c . Thus, e^{t^2} is not of exponential order.
 46. Assuming that (c) of Theorem 4.1 is applicable with a complex exponent, we have

$$\mathcal{L}\{e^{(a+ib)t}\} = \frac{1}{s - (a+ib)} = \frac{1}{(s-a) - ib} \frac{(s-a) + ib}{(s-a) + ib} = \frac{s-a+ib}{(s-a)^2 + b^2}.$$

By Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, so

$$\begin{aligned} \mathcal{L}\{e^{(a+ib)t}\} &= \mathcal{L}\{e^{at} e^{ibt}\} = \mathcal{L}\{e^{at} (\cos bt + i \sin bt)\} \\ &= \mathcal{L}\{e^{at} \cos bt\} + i \mathcal{L}\{e^{at} \sin bt\} \\ &= \frac{s-a}{(s-a)^2 + b^2} + i \frac{b}{(s-a)^2 + b^2}. \end{aligned}$$

Equating real and imaginary parts we get

$$\mathcal{L}\{e^{at} \cos bt\} = \frac{s-a}{(s-a)^2 + b^2} \quad \text{and} \quad \mathcal{L}\{e^{at} \sin bt\} = \frac{b}{(s-a)^2 + b^2}.$$

47. We want $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$ or

$$m(\alpha x + \beta y) + b = \alpha(mx + b) + \beta(my + b) = m(\alpha x + \beta y) + (\alpha + \beta)b$$

for all real numbers α and β . Taking $\alpha = \beta = 1$ we see that $b = 2b$, so $b = 0$. Thus, $f(x) = mx + b$ will be a linear transformation when $b = 0$.

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48. Assume that $\mathcal{L}\{t^{n-1}\} = (n-1)!/s^n$. Then, using the definition of the Laplace transform and integration by parts, we have

$$\begin{aligned}\mathcal{L}\{t^n\} &= \int_0^\infty e^{-st} t^n dt = -\frac{1}{s} e^{-st} t^n \Big|_0^\infty + \frac{n}{s} \int_0^\infty e^{-st} t^{n-1} dt \\ &= 0 + \frac{n}{s} \mathcal{L}\{t^{n-1}\} = \frac{n}{s} \frac{(n-1)!}{s^n} = \frac{n!}{s^{n+1}}.\end{aligned}$$

EXERCISES 4.2

The Inverse Transform and Transforms of Derivatives

1. $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\} = \frac{1}{2} t^2$
2. $\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} = \frac{1}{6} \mathcal{L}^{-1}\left\{\frac{3!}{s^4}\right\} = \frac{1}{6} t^3$
3. $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{48}{s^5}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{48}{24} \cdot \frac{4!}{s^5}\right\} = t - 2t^4$
4. $\mathcal{L}^{-1}\left\{\left(\frac{2}{s} - \frac{1}{s^3}\right)^2\right\} = \mathcal{L}^{-1}\left\{4 \cdot \frac{1}{s^2} - \frac{4}{6} \cdot \frac{3!}{s^4} + \frac{1}{120} \cdot \frac{5!}{s^6}\right\} = 4t - \frac{2}{3} t^3 + \frac{1}{120} t^5$
5. $\mathcal{L}^{-1}\left\{\frac{(s+1)^3}{s^4}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s} + 3 \cdot \frac{1}{s^2} + \frac{3}{2} \cdot \frac{2}{s^3} + \frac{1}{6} \cdot \frac{3!}{s^4}\right\} = 1 + 3t + \frac{3}{2} t^2 + \frac{1}{6} t^3$
6. $\mathcal{L}^{-1}\left\{\frac{(s+2)^2}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s} + 4 \cdot \frac{1}{s^2} + 2 \cdot \frac{2}{s^3}\right\} = 1 + 4t + 2t^2$
7. $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s-2}\right\} = t - 1 + e^{2t}$
8. $\mathcal{L}^{-1}\left\{\frac{4}{s} + \frac{6}{s^5} - \frac{1}{s+8}\right\} = \mathcal{L}^{-1}\left\{4 \cdot \frac{1}{s} + \frac{1}{4} \cdot \frac{4!}{s^5} - \frac{1}{s+8}\right\} = 4 + \frac{1}{4} t^4 - e^{-8t}$
9. $\mathcal{L}^{-1}\left\{\frac{1}{4s+1}\right\} = \frac{1}{4} \mathcal{L}^{-1}\left\{\frac{1}{s+1/4}\right\} = \frac{1}{4} e^{-t/4}$
10. $\mathcal{L}^{-1}\left\{\frac{1}{5s-2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{5} \cdot \frac{1}{s-2/5}\right\} = \frac{1}{5} e^{2t/5}$
11. $\mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{7} \cdot \frac{7}{s^2+49}\right\} = \frac{5}{7} \sin 7t$
12. $\mathcal{L}^{-1}\left\{\frac{10s}{s^2+16}\right\} = 10 \cos 4t$
13. $\mathcal{L}^{-1}\left\{\frac{4s}{4s^2+1}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+1/4}\right\} = \cos \frac{1}{2} t$
14. $\mathcal{L}^{-1}\left\{\frac{1}{4s^2+1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \cdot \frac{1/2}{s^2+1/4}\right\} = \frac{1}{2} \sin \frac{1}{2} t$

15. $\mathcal{L}^{-1}\left\{\frac{2s-6}{s^2+9}\right\} = \mathcal{L}^{-1}\left\{2 \cdot \frac{s}{s^2+9} - 2 \cdot \frac{3}{s^2+9}\right\} = 2 \cos 3t - 2 \sin 3t$
16. $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+2} + \frac{1}{\sqrt{2}} \frac{\sqrt{2}}{s^2+2}\right\} = \cos \sqrt{2}t + \frac{\sqrt{2}}{2} \sin \sqrt{2}t$
17. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+3s}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{3} \cdot \frac{1}{s} - \frac{1}{3} \cdot \frac{1}{s+3}\right\} = \frac{1}{3} - \frac{1}{3}e^{-3t}$
18. $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2-4s}\right\} = \mathcal{L}^{-1}\left\{-\frac{1}{4} \cdot \frac{1}{s} + \frac{5}{4} \cdot \frac{1}{s-4}\right\} = -\frac{1}{4} + \frac{5}{4}e^{4t}$
19. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+2s-3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{4} \cdot \frac{1}{s-1} + \frac{3}{4} \cdot \frac{1}{s+3}\right\} = \frac{1}{4}e^t + \frac{3}{4}e^{-3t}$
20. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+s-20}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{9} \cdot \frac{1}{s-4} - \frac{1}{9} \cdot \frac{1}{s+5}\right\} = \frac{1}{9}e^{4t} - \frac{1}{9}e^{-5t}$
21. $\mathcal{L}^{-1}\left\{\frac{0.9s}{(s-0.1)(s+0.2)}\right\} = \mathcal{L}^{-1}\left\{(0.3) \cdot \frac{1}{s-0.1} + (0.6) \cdot \frac{1}{s+0.2}\right\} = 0.3e^{0.1t} + 0.6e^{-0.2t}$
22. $\mathcal{L}^{-1}\left\{\frac{s-3}{(s-\sqrt{3})(s+\sqrt{3})}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2-3} - \sqrt{3} \cdot \frac{\sqrt{3}}{s^2-3}\right\} = \cosh \sqrt{3}t - \sqrt{3} \sinh \sqrt{3}t$
23. $\mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s-3)(s-6)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \cdot \frac{1}{s-2} - \frac{1}{s-3} + \frac{1}{2} \cdot \frac{1}{s-6}\right\} = \frac{1}{2}e^{2t} - e^{3t} + \frac{1}{2}e^{6t}$
24. $\mathcal{L}^{-1}\left\{\frac{s^2+1}{s(s-1)(s+1)(s-2)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \cdot \frac{1}{s} - \frac{1}{s-1} - \frac{1}{3} \cdot \frac{1}{s+1} + \frac{5}{6} \cdot \frac{1}{s-2}\right\}$
 $= \frac{1}{2} - e^t - \frac{1}{3}e^{-t} + \frac{5}{6}e^{2t}$
25. $\mathcal{L}^{-1}\left\{\frac{1}{s^3+5s}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s(s^2+5)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{5} \cdot \frac{1}{s} - \frac{1}{5} \frac{s}{s^2+5}\right\} = \frac{1}{5} - \frac{1}{5} \cos \sqrt{5}t$
26. $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+4)(s+2)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{4} \cdot \frac{s}{s^2+4} + \frac{1}{4} \cdot \frac{2}{s^2+4} - \frac{1}{4} \cdot \frac{1}{s+2}\right\} = \frac{1}{4} \cos 2t + \frac{1}{4} \sin 2t - \frac{1}{4}e^{-2t}$
27. $\mathcal{L}^{-1}\left\{\frac{2s-4}{(s^2+s)(s^2+1)}\right\} = \mathcal{L}^{-1}\left\{\frac{2s-4}{s(s+1)(s^2+1)}\right\} = \mathcal{L}^{-1}\left\{-\frac{4}{s} + \frac{3}{s+1} + \frac{s}{s^2+1} + \frac{3}{s^2+1}\right\}$
 $= -4 + 3e^{-t} + \cos t + 3 \sin t$
28. $\mathcal{L}^{-1}\left\{\frac{1}{s^4-9}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{6\sqrt{3}} \cdot \frac{\sqrt{3}}{s^2-3} - \frac{1}{6\sqrt{3}} \cdot \frac{\sqrt{3}}{s^2+3}\right\} = \frac{1}{6\sqrt{3}} \sinh \sqrt{3}t - \frac{1}{6\sqrt{3}} \sin \sqrt{3}t$
29. $\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{3} \cdot \frac{1}{s^2+1} - \frac{1}{3} \cdot \frac{1}{s^2+4}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{3} \cdot \frac{1}{s^2+1} - \frac{1}{6} \cdot \frac{2}{s^2+4}\right\}$
 $= \frac{1}{3} \sin t - \frac{1}{6} \sin 2t$
30. $\mathcal{L}^{-1}\left\{\frac{6s+3}{(s^2+1)(s^2+4)}\right\} = \mathcal{L}^{-1}\left\{2 \cdot \frac{s}{s^2+1} + \frac{1}{s^2+1} - 2 \cdot \frac{s}{s^2+4} - \frac{1}{2} \cdot \frac{2}{s^2+4}\right\}$
 $= 2 \cos t + \sin t - 2 \cos 2t - \frac{1}{2} \sin 2t$
31. The Laplace transform of the initial-value problem is

$$s \mathcal{L}\{y\} - y(0) - \mathcal{L}\{y\} = \frac{1}{s}.$$

4.2 The Inverse Transform and Transforms of Derivatives

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = -\frac{1}{s} + \frac{1}{s-1}.$$

Thus

$$y = -1 + e^t.$$

32. The Laplace transform of the initial-value problem is

$$2s\mathcal{L}\{y\} - 2y(0) + \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{6}{2s+1} = \frac{3}{s+1/2}.$$

Thus

$$y = 3e^{-t/2}.$$

33. The Laplace transform of the initial-value problem is

$$s\mathcal{L}\{y\} - y(0) + 6\mathcal{L}\{y\} = \frac{1}{s-4}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{(s-4)(s+6)} + \frac{2}{s+6} = \frac{1}{10} \cdot \frac{1}{s-4} + \frac{19}{10} \cdot \frac{1}{s+6}.$$

Thus

$$y = \frac{1}{10}e^{4t} + \frac{19}{10}e^{-6t}.$$

34. The Laplace transform of the initial-value problem is

$$s\mathcal{L}\{y\} - \mathcal{L}\{y\} = \frac{2s}{s^2+25}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2s}{(s-1)(s^2+25)} = \frac{1}{13} \cdot \frac{1}{s-1} - \frac{1}{13} \cdot \frac{s}{s^2+25} + \frac{5}{13} \cdot \frac{5}{s^2+25}.$$

Thus

$$y = \frac{1}{13}e^t - \frac{1}{13}\cos 5t + \frac{5}{13}\sin 5t.$$

35. The Laplace transform of the initial-value problem is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) + 5[s\mathcal{L}\{y\} - y(0)] + 4\mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s+5}{s^2+5s+4} = \frac{4}{3} \cdot \frac{1}{s+1} - \frac{1}{3} \cdot \frac{1}{s+4}.$$

Thus

$$y = \frac{4}{3}e^{-t} - \frac{1}{3}e^{-4t}.$$

36. The Laplace transform of the initial-value problem is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) - 4[s\mathcal{L}\{y\} - y(0)] = \frac{6}{s-3} - \frac{3}{s+1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{6}{(s-3)(s^2-4s)} - \frac{3}{(s+1)(s^2-4s)} + \frac{s-5}{s^2-4s} \\ &= \frac{5}{2} \cdot \frac{1}{s} - \frac{2}{s-3} - \frac{3}{5} \cdot \frac{1}{s+1} + \frac{11}{10} \cdot \frac{1}{s-4}.\end{aligned}$$

Thus

$$y = \frac{5}{2} - 2e^{3t} - \frac{3}{5}e^{-t} + \frac{11}{10}e^{4t}.$$

37. The Laplace transform of the initial-value problem is

$$s^2 \mathcal{L}\{y\} - sy(0) + \mathcal{L}\{y\} = \frac{2}{s^2 + 2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2}{(s^2+1)(s^2+2)} + \frac{10s}{s^2+1} = \frac{10s}{s^2+1} + \frac{2}{s^2+1} - \frac{2}{s^2+2}.$$

Thus

$$y = 10 \cos t + 2 \sin t - \sqrt{2} \sin \sqrt{2} t.$$

38. The Laplace transform of the initial-value problem is

$$s^2 \mathcal{L}\{y\} + 9 \mathcal{L}\{y\} = \frac{1}{s-1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{(s-1)(s^2+9)} = \frac{1}{10} \cdot \frac{1}{s-1} - \frac{1}{10} \cdot \frac{1}{s^2+9} - \frac{1}{10} \cdot \frac{s}{s^2+9}.$$

Thus

$$y = \frac{1}{10}e^t - \frac{1}{30} \sin 3t - \frac{1}{10} \cos 3t.$$

39. The Laplace transform of the initial-value problem is

$$2[s^3 \mathcal{L}\{y\} - s^2(0) - sy'(0) - y''(0)] + 3[s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] - 3[s \mathcal{L}\{y\} - y(0)] - 2 \mathcal{L}\{y\} = \frac{1}{s+1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2s+3}{(s+1)(s-1)(2s+1)(s+2)} = \frac{1}{2} \frac{1}{s+1} + \frac{5}{18} \frac{1}{s-1} - \frac{8}{9} \frac{1}{s+1/2} + \frac{1}{9} \frac{1}{s+2}.$$

Thus

$$y = \frac{1}{2}e^{-t} + \frac{5}{18}e^t - \frac{8}{9}e^{-t/2} + \frac{1}{9}e^{-2t}.$$

40. The Laplace transform of the initial-value problem is

$$s^3 \mathcal{L}\{y\} - s^2(0) - sy'(0) - y''(0) + 2[s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] - [s \mathcal{L}\{y\} - y(0)] - 2 \mathcal{L}\{y\} = \frac{3}{s^2+9}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{s^2+12}{(s-1)(s+1)(s+2)(s^2+9)} \\ &= \frac{13}{60} \frac{1}{s-1} - \frac{13}{20} \frac{1}{s+1} + \frac{16}{39} \frac{1}{s+2} + \frac{3}{130} \frac{s}{s^2+9} - \frac{1}{65} \frac{3}{s^2+9}.\end{aligned}$$

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Thus

$$y = \frac{13}{60}e^t - \frac{13}{20}e^{-t} + \frac{16}{39}e^{-2t} + \frac{3}{130}\cos 3t - \frac{1}{65}\sin 3t.$$

41. The Laplace transform of the initial-value problem is

$$s\mathcal{L}\{y\} + \mathcal{L}\{y\} = \frac{s+3}{s^2+6s+13}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{s+3}{(s+1)(s^2+6s+13)} = \frac{1}{4} \cdot \frac{1}{s+1} - \frac{1}{4} \cdot \frac{s+1}{s^2+6s+13} \\ &= \frac{1}{4} \cdot \frac{1}{s+1} - \frac{1}{4} \left(\frac{s+3}{(s+3)^2+4} - \frac{2}{(s+3)^2+4} \right).\end{aligned}$$

Thus

$$y = \frac{1}{4}e^{-t} - \frac{1}{4}e^{-3t}\cos 2t + \frac{1}{4}e^{-3t}\sin 2t.$$

42. The Laplace transform of the initial-value problem is

$$s^2\mathcal{L}\{y\} - s \cdot 1 - 3 - 2[s\mathcal{L}\{y\} - 1] + 5\mathcal{L}\{y\} = (s^2 - 2s + 5)\mathcal{L}\{y\} - s - 1 = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s+1}{s^2-2s+5} = \frac{s-1+2}{(s-1)^2+2^2} = \frac{s-1}{(s-1)^2+2^2} + \frac{2}{(s-1)^2+2^2}.$$

Thus

$$y = e^t \cos 2t + e^t \sin 2t.$$

43. (a) Differentiating $f(t) = te^{at}$ we get $f'(t) = ate^{at} + e^{at}$ so $\mathcal{L}\{ate^{at} + e^{at}\} = s\mathcal{L}\{te^{at}\}$, where we have used $f(0) = 0$. Writing the equation as

$$a\mathcal{L}\{te^{at}\} + \mathcal{L}\{e^{at}\} = s\mathcal{L}\{te^{at}\}$$

and solving for $\mathcal{L}\{te^{at}\}$ we get

$$\mathcal{L}\{te^{at}\} = \frac{1}{s-a}\mathcal{L}\{e^{at}\} = \frac{1}{(s-a)^2}.$$

(b) Starting with $f(t) = t \sin kt$ we have

$$f'(t) = kt \cos kt + \sin kt$$

$$f''(t) = -k^2t \sin kt + 2k \cos kt.$$

Then

$$\mathcal{L}\{-k^2t \sin kt + 2k \cos kt\} = s^2\mathcal{L}\{t \sin kt\}$$

where we have used $f(0) = 0$ and $f'(0) = 0$. Writing the above equation as

$$-k^2\mathcal{L}\{t \sin kt\} + 2k\mathcal{L}\{\cos kt\} = s^2\mathcal{L}\{t \sin kt\}$$

and solving for $\mathcal{L}\{t \sin kt\}$ gives

$$\mathcal{L}\{t \sin kt\} = \frac{2k}{s^2+k^2}\mathcal{L}\{\cos kt\} = \frac{2k}{s^2+k^2} \frac{s}{s^2+k^2} = \frac{2ks}{(s^2+k^2)^2}.$$

44. Let $f_1(t) = 1$ and $f_2(t) = \begin{cases} 1, & t \geq 0, \quad t \neq 1 \\ 0, & t = 1 \end{cases}$. Then $\mathcal{L}\{f_1(t)\} = \mathcal{L}\{f_2(t)\} = 1/s$, but $f_1(t) \neq f_2(t)$.

45. For $y'' - 4y' = 6e^{3t} - 3e^{-t}$ the transfer function is $W(s) = 1/(s^2 - 4s)$. The zero-input response is

$$y_0(t) = \mathcal{L}^{-1} \left\{ \frac{s-5}{s^2-4s} \right\} = \mathcal{L}^{-1} \left\{ \frac{5}{4} \cdot \frac{1}{s} - \frac{1}{4} \cdot \frac{1}{s-4} \right\} = \frac{5}{4} - \frac{1}{4}e^{4t},$$

and the zero-state response is

$$\begin{aligned} y_1(t) &= \mathcal{L}^{-1} \left\{ \frac{6}{(s-3)(s^2-4s)} - \frac{3}{(s+1)(s^2-4s)} \right\} \\ &= \mathcal{L}^{-1} \left\{ \frac{27}{20} \cdot \frac{1}{s-4} - \frac{2}{s-3} + \frac{5}{4} \cdot \frac{1}{s} - \frac{3}{5} \cdot \frac{1}{s+1} \right\} \\ &= \frac{27}{20}e^{4t} - 2e^{3t} + \frac{5}{4} - \frac{3}{5}e^{-t}. \end{aligned}$$

46. From Theorem 4.4, if f and f' are continuous and of exponential order, $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$. From Theorem 4.5, $\lim_{s \rightarrow \infty} \mathcal{L}\{f'(t)\} = 0$ so

$$\lim_{s \rightarrow \infty} [sF(s) - f(0)] = 0 \quad \text{and} \quad \lim_{s \rightarrow \infty} F(s) = f(0).$$

For $f(t) = \cos kt$,

$$\lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} s \frac{s}{s^2 + k^2} = 1 = f(0).$$

EXERCISES 4.3

Translation Theorems

$$1. \mathcal{L}\{te^{10t}\} = \frac{1}{(s-10)^2}$$

$$2. \mathcal{L}\{te^{-6t}\} = \frac{1}{(s+6)^2}$$

$$3. \mathcal{L}\{t^3e^{-2t}\} = \frac{3!}{(s+2)^4}$$

$$4. \mathcal{L}\{t^{10}e^{-7t}\} = \frac{10!}{(s+7)^{11}}$$

$$5. \mathcal{L}\{t(e^t + e^{2t})^2\} = \mathcal{L}\{te^{2t} + 2te^{3t} + te^{4t}\} = \frac{1}{(s-2)^2} + \frac{2}{(s-3)^2} + \frac{1}{(s-4)^2}$$

$$6. \mathcal{L}\{e^{2t}(t-1)^2\} = \mathcal{L}\{t^2e^{2t} - 2te^{2t} + e^{2t}\} = \frac{2}{(s-2)^3} - \frac{2}{(s-2)^2} + \frac{1}{s-2}$$

$$7. \mathcal{L}\{e^t \sin 3t\} = \frac{3}{(s-1)^2 + 9}$$

$$8. \mathcal{L}\{e^{-2t} \cos 4t\} = \frac{s+2}{(s+2)^2 + 16}$$

$$9. \mathcal{L}\{(1 - e^t + 3e^{-4t}) \cos 5t\} = \mathcal{L}\{\cos 5t - e^t \cos 5t + 3e^{-4t} \cos 5t\} = \frac{s}{s^2 + 25} - \frac{s-1}{(s-1)^2 + 25} + \frac{3(s+4)}{(s+4)^2 + 25}$$

$$10. \mathcal{L}\left\{e^{3t} \left(9 - 4t + 10 \sin \frac{t}{2}\right)\right\} = \mathcal{L}\left\{9e^{3t} - 4te^{3t} + 10e^{3t} \sin \frac{t}{2}\right\} = \frac{9}{s-3} - \frac{4}{(s-3)^2} + \frac{5}{(s-3)^2 + 1/4}$$

$$11. \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \frac{2}{(s+2)^3}\right\} = \frac{1}{2}t^2e^{-2t}$$

4.3 Translation Theorems

12. $\mathcal{L}^{-1}\left\{\frac{1}{(s-1)^4}\right\} = \frac{1}{6}\mathcal{L}^{-1}\left\{\frac{3!}{(s-1)^4}\right\} = \frac{1}{6}t^3e^t$
13. $\mathcal{L}^{-1}\left\{\frac{1}{s^2-6s+10}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2+1^2}\right\} = e^{3t}\sin t$
14. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+2s+5}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2}\frac{2}{(s+1)^2+2^2}\right\} = \frac{1}{2}e^{-t}\sin 2t$
15. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+4s+5}\right\} = \mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+1^2} - 2\frac{1}{(s+2)^2+1^2}\right\} = e^{-2t}\cos t - 2e^{-2t}\sin t$
16. $\mathcal{L}^{-1}\left\{\frac{2s+5}{s^2+6s+34}\right\} = \mathcal{L}^{-1}\left\{2\frac{(s+3)}{(s+3)^2+5^2} - \frac{1}{5}\frac{5}{(s+3)^2+5^2}\right\} = 2e^{-3t}\cos 5t - \frac{1}{5}e^{-3t}\sin 5t$
17. $\mathcal{L}^{-1}\left\{\frac{s}{(s+1)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{s+1-1}{(s+1)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+1} - \frac{1}{(s+1)^2}\right\} = e^{-t} - te^{-t}$
18. $\mathcal{L}^{-1}\left\{\frac{5s}{(s-2)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{5(s-2)+10}{(s-2)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{s-2} + \frac{10}{(s-2)^2}\right\} = 5e^{2t} + 10te^{2t}$
19. $\mathcal{L}^{-1}\left\{\frac{2s-1}{s^2(s+1)^3}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{s} - \frac{1}{s^2} - \frac{5}{s+1} - \frac{4}{(s+1)^2} - \frac{3}{2}\frac{2}{(s+1)^3}\right\} = 5 - t - 5e^{-t} - 4te^{-t} - \frac{3}{2}t^2e^{-t}$
20. $\mathcal{L}^{-1}\left\{\frac{(s+1)^2}{(s+2)^4}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2} - \frac{2}{(s+2)^3} + \frac{1}{6}\frac{3!}{(s+2)^4}\right\} = te^{-2t} - t^2e^{-2t} + \frac{1}{6}t^3e^{-2t}$
21. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - y(0) + 4\mathcal{L}\{y\} = \frac{1}{s+4}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{(s+4)^2} + \frac{2}{s+4}.$$

Thus

$$y = te^{-4t} + 2e^{-4t}.$$

22. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - \mathcal{L}\{y\} = \frac{1}{s} + \frac{1}{(s-1)^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{s(s-1)} + \frac{1}{(s-1)^3} = -\frac{1}{s} + \frac{1}{s-1} + \frac{1}{(s-1)^3}.$$

Thus

$$y = -1 + e^t + \frac{1}{2}t^2e^t.$$

23. The Laplace transform of the differential equation is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) + 2[s\mathcal{L}\{y\} - y(0)] + \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s+3}{(s+1)^2} = \frac{1}{s+1} + \frac{2}{(s+1)^2}.$$

Thus

$$y = e^{-t} + 2te^{-t}.$$

24. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 4[s \mathcal{L}\{y\} - y(0)] + 4 \mathcal{L}\{y\} = \frac{6}{(s-2)^4}.$$

Solving for $\mathcal{L}\{y\}$ we obtain $\mathcal{L}\{y\} = \frac{1}{20} \frac{5!}{(s-2)^6}$. Thus, $y = \frac{1}{20} t^5 e^{2t}$.

25. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 6[s \mathcal{L}\{y\} - y(0)] + 9 \mathcal{L}\{y\} = \frac{1}{s^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1+s^2}{s^2(s-3)^2} = \frac{2}{27} \frac{1}{s} + \frac{1}{9} \frac{1}{s^2} - \frac{2}{27} \frac{1}{s-3} + \frac{10}{9} \frac{1}{(s-3)^2}.$$

Thus

$$y = \frac{2}{27} + \frac{1}{9}t - \frac{2}{27}e^{3t} + \frac{10}{9}te^{3t}.$$

26. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 4[s \mathcal{L}\{y\} - y(0)] + 4 \mathcal{L}\{y\} = \frac{6}{s^4}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s^5 - 4s^4 + 6}{s^4(s-2)^2} = \frac{3}{4} \frac{1}{s} + \frac{9}{8} \frac{1}{s^2} + \frac{3}{4} \frac{2}{s^3} + \frac{1}{4} \frac{3!}{s^4} + \frac{1}{4} \frac{1}{s-2} - \frac{13}{8} \frac{1}{(s-2)^2}.$$

Thus

$$y = \frac{3}{4} + \frac{9}{8}t + \frac{3}{4}t^2 + \frac{1}{4}t^3 + \frac{1}{4}e^{2t} - \frac{13}{8}te^{2t}.$$

27. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 6[s \mathcal{L}\{y\} - y(0)] + 13 \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = -\frac{3}{s^2 - 6s + 13} = -\frac{3}{2} \frac{2}{(s-3)^2 + 2^2}.$$

Thus

$$y = -\frac{3}{2}e^{3t} \sin 2t.$$

28. The Laplace transform of the differential equation is

$$2[s^2 \mathcal{L}\{y\} - sy(0)] + 20[s \mathcal{L}\{y\} - y(0)] + 51 \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{4s+40}{2s^2+20s+51} = \frac{2s+20}{(s+5)^2+1/2} = \frac{2(s+5)}{(s+5)^2+1/2} + \frac{10}{(s+5)^2+1/2}.$$

Thus

$$y = 2e^{-5t} \cos(t/\sqrt{2}) + 10\sqrt{2}e^{-5t} \sin(t/\sqrt{2}).$$

29. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - [s \mathcal{L}\{y\} - y(0)] = \frac{s-1}{(s-1)^2+1}.$$

4.3 Translation Theorems

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{s(s^2 - 2s + 2)} = \frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{s-1}{(s-1)^2 + 1} + \frac{1}{2} \frac{1}{(s-1)^2 + 1}.$$

Thus

$$y = \frac{1}{2} - \frac{1}{2}e^t \cos t + \frac{1}{2}e^t \sin t.$$

30. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 2[s \mathcal{L}\{y\} - y(0)] + 5 \mathcal{L}\{y\} = \frac{1}{s} + \frac{1}{s^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} &= \frac{4s^2 + s + 1}{s^2(s^2 - 2s + 5)} = \frac{7}{25} \frac{1}{s} + \frac{1}{5} \frac{1}{s^2} + \frac{-7s/25 + 109/25}{s^2 - 2s + 5} \\ &= \frac{7}{25} \frac{1}{s} + \frac{1}{5} \frac{1}{s^2} - \frac{7}{25} \frac{s-1}{(s-1)^2 + 2^2} + \frac{51}{25} \frac{2}{(s-1)^2 + 2^2}. \end{aligned}$$

Thus

$$y = \frac{7}{25} + \frac{1}{5}t - \frac{7}{25}e^t \cos 2t + \frac{51}{25}e^t \sin 2t.$$

31. Taking the Laplace transform of both sides of the differential equation and letting $c = y(0)$ we obtain

$$\begin{aligned} \mathcal{L}\{y''\} + \mathcal{L}\{2y'\} + \mathcal{L}\{y\} &= 0 \\ s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 2s \mathcal{L}\{y\} - 2y(0) + \mathcal{L}\{y\} &= 0 \\ s^2 \mathcal{L}\{y\} - cs - 2 + 2s \mathcal{L}\{y\} - 2c + \mathcal{L}\{y\} &= 0 \\ (s^2 + 2s + 1) \mathcal{L}\{y\} &= cs + 2c + 2 \\ \mathcal{L}\{y\} &= \frac{cs}{(s+1)^2} + \frac{2c+2}{(s+1)^2} \\ &= c \frac{s+1-1}{(s+1)^2} + \frac{2c+2}{(s+1)^2} \\ &= \frac{c}{s+1} + \frac{c+2}{(s+1)^2}. \end{aligned}$$

Therefore,

$$y(t) = c \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} + (c+2) \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2} \right\} = ce^{-t} + (c+2)te^{-t}.$$

To find c we let $y(1) = 2$. Then $2 = ce^{-1} + (c+2)e^{-1} = 2(c+1)e^{-1}$ and $c = e - 1$. Thus

$$y(t) = (e-1)e^{-t} + (e+1)te^{-t}.$$

32. Taking the Laplace transform of both sides of the differential equation and letting $c = y'(0)$ we obtain

$$\begin{aligned} \mathcal{L}\{y''\} + \mathcal{L}\{8y'\} + \mathcal{L}\{20y\} &= 0 \\ s^2 \mathcal{L}\{y\} - y'(0) + 8s \mathcal{L}\{y\} + 20 \mathcal{L}\{y\} &= 0 \\ s^2 \mathcal{L}\{y\} - c + 8s \mathcal{L}\{y\} + 20 \mathcal{L}\{y\} &= 0 \\ (s^2 + 8s + 20) \mathcal{L}\{y\} &= c \\ \mathcal{L}\{y\} &= \frac{c}{s^2 + 8s + 20} = \frac{c}{(s+4)^2 + 4}. \end{aligned}$$

Therefore,

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{c}{(s+4)^2 + 4} \right\} = \frac{c}{2} e^{-4t} \sin 2t = c_1 e^{-4t} \sin 2t.$$

To find c we let $y'(\pi) = 0$. Then $0 = y'(\pi) = ce^{-4\pi}$ and $c = 0$. Thus, $y(t) = 0$. (Since the differential equation is homogeneous and both boundary conditions are 0, we can see immediately that $y(t) = 0$ is a solution. We have shown that it is the only solution.)

- 33.** Recall from Section 3.8 that $mx'' = -kx - \beta x'$. Now $m = W/g = 4/32 = \frac{1}{8}$ slug, and $4 = 2k$ so that $k = 2$ lb/ft. Thus, the differential equation is $x'' + 7x' + 16x = 0$. The initial conditions are $x(0) = -3/2$ and $x'(0) = 0$. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{x\} + \frac{3}{2}s + 7s \mathcal{L}\{x\} + \frac{21}{2} + 16 \mathcal{L}\{x\} = 0.$$

Solving for $\mathcal{L}\{x\}$ we obtain

$$\mathcal{L}\{x\} = \frac{-3s/2 - 21/2}{s^2 + 7s + 16} = -\frac{3}{2} \frac{s + 7/2}{(s + 7/2)^2 + (\sqrt{15}/2)^2} - \frac{7\sqrt{15}}{10} \frac{\sqrt{15}/2}{(s + 7/2)^2 + (\sqrt{15}/2)^2}.$$

Thus

$$x = -\frac{3}{2} e^{-7t/2} \cos \frac{\sqrt{15}}{2} t - \frac{7\sqrt{15}}{10} e^{-7t/2} \sin \frac{\sqrt{15}}{2} t.$$

- 34.** The differential equation is

$$\frac{d^2 q}{dt^2} + 20 \frac{dq}{dt} + 200q = 150, \quad q(0) = q'(0) = 0.$$

The Laplace transform of this equation is

$$s^2 \mathcal{L}\{q\} + 20s \mathcal{L}\{q\} + 200 \mathcal{L}\{q\} = \frac{150}{s}.$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \frac{150}{s(s^2 + 20s + 200)} = \frac{3}{4} \frac{1}{s} - \frac{3}{4} \frac{s + 10}{(s + 10)^2 + 10^2} - \frac{3}{4} \frac{10}{(s + 10)^2 + 10^2}.$$

Thus

$$q(t) = \frac{3}{4} - \frac{3}{4} e^{-10t} \cos 10t - \frac{3}{4} e^{-10t} \sin 10t$$

and

$$i(t) = q'(t) = 15e^{-10t} \sin 10t.$$

- 35.** The differential equation is

$$\frac{d^2 q}{dt^2} + 2\lambda \frac{dq}{dt} + \omega^2 q = \frac{E_0}{L}, \quad q(0) = q'(0) = 0.$$

The Laplace transform of this equation is

$$s^2 \mathcal{L}\{q\} + 2\lambda s \mathcal{L}\{q\} + \omega^2 \mathcal{L}\{q\} = \frac{E_0}{L} \frac{1}{s}$$

or

$$(s^2 + 2\lambda s + \omega^2) \mathcal{L}\{q\} = \frac{E_0}{L} \frac{1}{s}.$$

Solving for $\mathcal{L}\{q\}$ and using partial fractions we obtain

$$\mathcal{L}\{q\} = \frac{E_0}{L} \left(\frac{1/\omega^2}{s} - \frac{(1/\omega^2)s + 2\lambda/\omega^2}{s^2 + 2\lambda s + \omega^2} \right) = \frac{E_0}{L\omega^2} \left(\frac{1}{s} - \frac{s + 2\lambda}{s^2 + 2\lambda s + \omega^2} \right).$$

4.3 Translation Theorems

For $\lambda > \omega$ we write $s^2 + 2\lambda s + \omega^2 = (s + \lambda)^2 - (\lambda^2 - \omega^2)$, so (recalling that $\omega^2 = 1/LC$)

$$\mathcal{L}\{q\} = E_0 C \left(\frac{1}{s} - \frac{s + \lambda}{(s + \lambda)^2 - (\lambda^2 - \omega^2)} - \frac{\lambda}{(s + \lambda)^2 - (\lambda^2 - \omega^2)} \right).$$

Thus for $\lambda > \omega$,

$$q(t) = E_0 C \left[1 - e^{-\lambda t} \left(\cosh \sqrt{\lambda^2 - \omega^2} t - \frac{\lambda}{\sqrt{\lambda^2 - \omega^2}} \sinh \sqrt{\lambda^2 - \omega^2} t \right) \right].$$

For $\lambda < \omega$ we write $s^2 + 2\lambda s + \omega^2 = (s + \lambda)^2 + (\omega^2 - \lambda^2)$, so

$$\mathcal{L}\{q\} = E_0 C \left(\frac{1}{s} - \frac{s + \lambda}{(s + \lambda)^2 + (\omega^2 - \lambda^2)} - \frac{\lambda}{(s + \lambda)^2 + (\omega^2 - \lambda^2)} \right).$$

Thus for $\lambda < \omega$,

$$q(t) = E_0 C \left[1 - e^{-\lambda t} \left(\cos \sqrt{\omega^2 - \lambda^2} t - \frac{\lambda}{\sqrt{\omega^2 - \lambda^2}} \sin \sqrt{\omega^2 - \lambda^2} t \right) \right].$$

For $\lambda = \omega$, $s^2 + 2\lambda s + \omega^2 = (s + \lambda)^2$ and

$$\mathcal{L}\{q\} = \frac{E_0}{L} \frac{1}{s(s + \lambda)^2} = \frac{E_0}{L} \left(\frac{1/\lambda^2}{s} - \frac{1/\lambda^2}{s + \lambda} - \frac{1/\lambda}{(s + \lambda)^2} \right) = \frac{E_0}{L\lambda^2} \left(\frac{1}{s} - \frac{1}{s + \lambda} - \frac{\lambda}{(s + \lambda)^2} \right).$$

Thus for $\lambda = \omega$,

$$q(t) = E_0 C (1 - e^{-\lambda t} - \lambda t e^{-\lambda t}).$$

36. The differential equation is

$$R \frac{dq}{dt} + \frac{1}{C} q = E_0 e^{-kt}, \quad q(0) = 0.$$

The Laplace transform of this equation is

$$Rs \mathcal{L}\{q\} + \frac{1}{C} \mathcal{L}\{q\} = E_0 \frac{1}{s + k}.$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \frac{E_0 C}{(s + k)(RCs + 1)} = \frac{E_0/R}{(s + k)(s + 1/RC)}.$$

When $1/RC \neq k$ we have by partial fractions

$$\mathcal{L}\{q\} = \frac{E_0}{R} \left(\frac{1/(1/RC - k)}{s + k} - \frac{1/(1/RC - k)}{s + 1/RC} \right) = \frac{E_0}{R} \frac{1}{1/RC - k} \left(\frac{1}{s + k} - \frac{1}{s + 1/RC} \right).$$

Thus

$$q(t) = \frac{E_0 C}{1 - kRC} \left(e^{-kt} - e^{-t/RC} \right).$$

When $1/RC = k$ we have

$$\mathcal{L}\{q\} = \frac{E_0}{R} \frac{1}{(s + k)^2}.$$

Thus

$$q(t) = \frac{E_0}{R} t e^{-kt} = \frac{E_0}{R} t e^{-t/RC}.$$

37. $\mathcal{L}\{(t - 1) \mathcal{U}(t - 1)\} = \frac{e^{-s}}{s^2}$

38. $\mathcal{L}\{e^{2-t} \mathcal{U}(t - 2)\} = \mathcal{L}\{e^{-(t-2)} \mathcal{U}(t - 2)\} = \frac{e^{-2s}}{s + 1}$

$$39. \mathcal{L}\{t \mathcal{U}(t-2)\} = \mathcal{L}\{(t-2) \mathcal{U}(t-2) + 2 \mathcal{U}(t-2)\} = \frac{e^{-2s}}{s^2} + \frac{2e^{-2s}}{s}$$

Alternatively, (16) of this section could be used:

$$\mathcal{L}\{t \mathcal{U}(t-2)\} = e^{-2s} \mathcal{L}\{t+2\} = e^{-2s} \left(\frac{1}{s^2} + \frac{2}{s} \right).$$

$$40. \mathcal{L}\{(3t+1) \mathcal{U}(t-1)\} = 3 \mathcal{L}\{(t-1) \mathcal{U}(t-1)\} + 4 \mathcal{L}\{\mathcal{U}(t-1)\} = \frac{3e^{-s}}{s^2} + \frac{4e^{-s}}{s}$$

Alternatively, (16) of this section could be used:

$$\mathcal{L}\{(3t+1) \mathcal{U}(t-1)\} = e^{-s} \mathcal{L}\{3t+4\} = e^{-s} \left(\frac{3}{s^2} + \frac{4}{s} \right).$$

$$41. \mathcal{L}\{\cos 2t \mathcal{U}(t-\pi)\} = \mathcal{L}\{\cos 2(t-\pi) \mathcal{U}(t-\pi)\} = \frac{se^{-\pi s}}{s^2+4}$$

Alternatively, (16) of this section could be used:

$$\mathcal{L}\{\cos 2t \mathcal{U}(t-\pi)\} = e^{-\pi s} \mathcal{L}\{\cos 2(t+\pi)\} = e^{-\pi s} \mathcal{L}\{\cos 2t\} = e^{-\pi s} \frac{s}{s^2+4}.$$

$$42. \mathcal{L}\left\{\sin t \mathcal{U}\left(t - \frac{\pi}{2}\right)\right\} = \mathcal{L}\left\{\cos\left(t - \frac{\pi}{2}\right) \mathcal{U}\left(t - \frac{\pi}{2}\right)\right\} = \frac{se^{-\pi s/2}}{s^2+1}$$

Alternatively, (16) of this section could be used:

$$\mathcal{L}\left\{\sin t \mathcal{U}\left(t - \frac{\pi}{2}\right)\right\} = e^{-\pi s/2} \mathcal{L}\left\{\sin\left(t + \frac{\pi}{2}\right)\right\} = e^{-\pi s/2} \mathcal{L}\{\cos t\} = e^{-\pi s/2} \frac{s}{s^2+1}.$$

$$43. \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \cdot \frac{2}{s^3} e^{-2s}\right\} = \frac{1}{2}(t-2)^2 \mathcal{U}(t-2)$$

$$44. \mathcal{L}^{-1}\left\{\frac{(1+e^{-2s})^2}{s+2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+2} + \frac{2e^{-2s}}{s+2} + \frac{e^{-4s}}{s+2}\right\} = e^{-2t} + 2e^{-2(t-2)} \mathcal{U}(t-2) + e^{-2(t-4)} \mathcal{U}(t-4)$$

$$45. \mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1}\right\} = \sin(t-\pi) \mathcal{U}(t-\pi) = -\sin t \mathcal{U}(t-\pi)$$

$$46. \mathcal{L}^{-1}\left\{\frac{se^{-\pi s/2}}{s^2+4}\right\} = \cos 2\left(t - \frac{\pi}{2}\right) \mathcal{U}\left(t - \frac{\pi}{2}\right) = -\cos 2t \mathcal{U}\left(t - \frac{\pi}{2}\right)$$

$$47. \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s(s+1)}\right\} = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s} - \frac{e^{-s}}{s+1}\right\} = \mathcal{U}(t-1) - e^{-(t-1)} \mathcal{U}(t-1)$$

$$48. \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2(s-1)}\right\} = \mathcal{L}^{-1}\left\{-\frac{e^{-2s}}{s} - \frac{e^{-2s}}{s^2} + \frac{e^{-2s}}{s-1}\right\} = -\mathcal{U}(t-2) - (t-2) \mathcal{U}(t-2) + e^{t-2} \mathcal{U}(t-2)$$

$$49. \text{ (c)} \qquad 50. \text{ (e)} \qquad 51. \text{ (f)} \qquad 52. \text{ (b)} \qquad 53. \text{ (a)} \qquad 54. \text{ (d)}$$

$$55. \mathcal{L}\{2-4 \mathcal{U}(t-3)\} = \frac{2}{s} - \frac{4}{s} e^{-3s}$$

$$56. \mathcal{L}\{1 - \mathcal{U}(t-4) + \mathcal{U}(t-5)\} = \frac{1}{s} - \frac{e^{-4s}}{s} + \frac{e^{-5s}}{s}$$

$$57. \mathcal{L}\{t^2 \mathcal{U}(t-1)\} = \mathcal{L}\{[(t-1)^2 + 2(t-1) - 1] \mathcal{U}(t-1)\} = \mathcal{L}\{[(t-1)^2 + 2(t-1) - 1] \mathcal{U}(t-1)\} \\ = \left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} \right) e^{-s}$$

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Alternatively, by (16) of this section,

$$\mathcal{L}\{t^2 \mathcal{U}(t-1)\} = e^{-s} \mathcal{L}\{t^2 + 2t + 1\} = e^{-s} \left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} \right).$$

$$58. \quad \mathcal{L}\left\{\sin t \mathcal{U}\left(t - \frac{3\pi}{2}\right)\right\} = \mathcal{L}\left\{-\cos\left(t - \frac{3\pi}{2}\right) \mathcal{U}\left(t - \frac{3\pi}{2}\right)\right\} = -\frac{se^{-3\pi/2}}{s^2 + 1}$$

$$59. \quad \mathcal{L}\{t - t \mathcal{U}(t-2)\} = \mathcal{L}\{t - (t-2) \mathcal{U}(t-2) - 2 \mathcal{U}(t-2)\} = \frac{1}{s^2} - \frac{e^{-2s}}{s^2} - \frac{2e^{-2s}}{s}$$

$$60. \quad \mathcal{L}\{\sin t - \sin t \mathcal{U}(t-2\pi)\} = \mathcal{L}\{\sin t - \sin(t-2\pi) \mathcal{U}(t-2\pi)\} = \frac{1}{s^2 + 1} - \frac{e^{-2\pi s}}{s^2 + 1}$$

$$61. \quad \mathcal{L}\{f(t)\} = \mathcal{L}\{\mathcal{U}(t-a) - \mathcal{U}(t-b)\} = \frac{e^{-as}}{s} - \frac{e^{-bs}}{s}$$

$$62. \quad \mathcal{L}\{f(t)\} = \mathcal{L}\{\mathcal{U}(t-1) + \mathcal{U}(t-2) + \mathcal{U}(t-3) + \cdots\} = \frac{e^{-s}}{s} + \frac{e^{-2s}}{s} + \frac{e^{-3s}}{s} + \cdots = \frac{1}{s} \frac{e^{-s}}{1 - e^{-s}}$$

63. The Laplace transform of the differential equation is

$$s \mathcal{L}\{y\} - y(0) + \mathcal{L}\{y\} = \frac{5}{s} e^{-s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{5e^{-s}}{s(s+1)} = 5e^{-s} \left[\frac{1}{s} - \frac{1}{s+1} \right].$$

Thus

$$y = 5 \mathcal{U}(t-1) - 5e^{-(t-1)} \mathcal{U}(t-1).$$

64. The Laplace transform of the differential equation is

$$s \mathcal{L}\{y\} - y(0) + \mathcal{L}\{y\} = \frac{1}{s} - \frac{2}{s} e^{-s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{s(s+1)} - \frac{2e^{-s}}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1} - 2e^{-s} \left[\frac{1}{s} - \frac{1}{s+1} \right].$$

Thus

$$y = 1 - e^{-t} - 2 \left[1 - e^{-(t-1)} \right] \mathcal{U}(t-1).$$

65. The Laplace transform of the differential equation is

$$s \mathcal{L}\{y\} - y(0) + 2 \mathcal{L}\{y\} = \frac{1}{s^2} - e^{-s} \frac{s+1}{s^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{s^2(s+2)} - e^{-s} \frac{s+1}{s^2(s+2)} = -\frac{1}{4} \frac{1}{s} + \frac{1}{2} \frac{1}{s^2} + \frac{1}{4} \frac{1}{s+2} - e^{-s} \left[\frac{1}{4} \frac{1}{s} + \frac{1}{2} \frac{1}{s^2} - \frac{1}{4} \frac{1}{s+2} \right].$$

Thus

$$y = -\frac{1}{4} + \frac{1}{2}t + \frac{1}{4}e^{-2t} - \left[\frac{1}{4} + \frac{1}{2}(t-1) - \frac{1}{4}e^{-2(t-1)} \right] \mathcal{U}(t-1).$$

66. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 4 \mathcal{L}\{y\} = \frac{1}{s} - \frac{e^{-s}}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1-s}{s(s^2+4)} - e^{-s} \frac{1}{s(s^2+4)} = \frac{1}{4} \frac{1}{s} - \frac{1}{4} \frac{s}{s^2+4} - \frac{1}{2} \frac{2}{s^2+4} - e^{-s} \left[\frac{1}{4} \frac{1}{s} - \frac{1}{4} \frac{s}{s^2+4} \right].$$

Thus

$$y = \frac{1}{4} - \frac{1}{4} \cos 2t - \frac{1}{2} \sin 2t - \left[\frac{1}{4} - \frac{1}{4} \cos 2(t-1) \right] \mathcal{U}(t-1).$$

67. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 4 \mathcal{L}\{y\} = e^{-2\pi s} \frac{1}{s^2+1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s}{s^2+4} + e^{-2\pi s} \left[\frac{1}{3} \frac{1}{s^2+1} - \frac{1}{6} \frac{2}{s^2+4} \right].$$

Thus

$$y = \cos 2t + \left[\frac{1}{3} \sin(t-2\pi) - \frac{1}{6} \sin 2(t-2\pi) \right] \mathcal{U}(t-2\pi).$$

68. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 5[s \mathcal{L}\{y\} - y(0)] + 6 \mathcal{L}\{y\} = \frac{e^{-s}}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} &= e^{-s} \frac{1}{s(s-2)(s-3)} + \frac{1}{(s-2)(s-3)} \\ &= e^{-s} \left[\frac{1}{6} \frac{1}{s} - \frac{1}{2} \frac{1}{s-2} + \frac{1}{3} \frac{1}{s-3} \right] - \frac{1}{s-2} + \frac{1}{s-3}. \end{aligned}$$

Thus

$$y = \left[\frac{1}{6} - \frac{1}{2} e^{2(t-1)} + \frac{1}{3} e^{3(t-1)} \right] \mathcal{U}(t-1) - e^{2t} + e^{3t}.$$

69. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + \mathcal{L}\{y\} = \frac{e^{-\pi s}}{s} - \frac{e^{-2\pi s}}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = e^{-\pi s} \left[\frac{1}{s} - \frac{s}{s^2+1} \right] - e^{-2\pi s} \left[\frac{1}{s} - \frac{s}{s^2+1} \right] + \frac{1}{s^2+1}.$$

Thus

$$y = [1 - \cos(t-\pi)] \mathcal{U}(t-\pi) - [1 - \cos(t-2\pi)] \mathcal{U}(t-2\pi) + \sin t.$$

70. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 4[s \mathcal{L}\{y\} - y(0)] + 3 \mathcal{L}\{y\} = \frac{1}{s} - \frac{e^{-2s}}{s} - \frac{e^{-4s}}{s} + \frac{e^{-6s}}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} &= \frac{1}{3} \frac{1}{s} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{6} \frac{1}{s+3} - e^{-2s} \left[\frac{1}{3} \frac{1}{s} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{6} \frac{1}{s+3} \right] \\ &\quad - e^{-4s} \left[\frac{1}{3} \frac{1}{s} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{6} \frac{1}{s+3} \right] + e^{-6s} \left[\frac{1}{3} \frac{1}{s} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{6} \frac{1}{s+3} \right]. \end{aligned}$$

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Thus

$$y = \frac{1}{3} - \frac{1}{2}e^{-t} + \frac{1}{6}e^{-3t} - \left[\frac{1}{3} - \frac{1}{2}e^{-(t-2)} + \frac{1}{6}e^{-3(t-2)} \right] \mathcal{U}(t-2) \\ - \left[\frac{1}{3} - \frac{1}{2}e^{-(t-4)} + \frac{1}{6}e^{-3(t-4)} \right] \mathcal{U}(t-4) + \left[\frac{1}{3} - \frac{1}{2}e^{-(t-6)} + \frac{1}{6}e^{-3(t-6)} \right] \mathcal{U}(t-6).$$

- 71.** Recall from Section 3.8 that $mx'' = -kx + f(t)$. Now $m = W/g = 32/32 = 1$ slug, and $32 = 2k$ so that $k = 16$ lb/ft. Thus, the differential equation is $x'' + 16x = f(t)$. The initial conditions are $x(0) = 0$, $x'(0) = 0$. Also, since

$$f(t) = \begin{cases} 20t, & 0 \leq t < 5 \\ 0, & t \geq 5 \end{cases}$$

and $20t = 20(t-5) + 100$ we can write

$$f(t) = 20t - 20t \mathcal{U}(t-5) = 20t - 20(t-5) \mathcal{U}(t-5) - 100 \mathcal{U}(t-5).$$

The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{x\} + 16 \mathcal{L}\{x\} = \frac{20}{s^2} - \frac{20}{s^2} e^{-5s} - \frac{100}{s} e^{-5s}.$$

Solving for $\mathcal{L}\{x\}$ we obtain

$$\mathcal{L}\{x\} = \frac{20}{s^2(s^2+16)} - \frac{20}{s^2(s^2+16)} e^{-5s} - \frac{100}{s(s^2+16)} e^{-5s} \\ = \left(\frac{5}{4} \cdot \frac{1}{s^2} - \frac{5}{16} \cdot \frac{4}{s^2+16} \right) (1 - e^{-5s}) - \left(\frac{25}{4} \cdot \frac{1}{s} - \frac{25}{4} \cdot \frac{s}{s^2+16} \right) e^{-5s}.$$

Thus

$$x(t) = \frac{5}{4}t - \frac{5}{16} \sin 4t - \left[\frac{5}{4}(t-5) - \frac{5}{16} \sin 4(t-5) \right] \mathcal{U}(t-5) - \left[\frac{25}{4} - \frac{25}{4} \cos 4(t-5) \right] \mathcal{U}(t-5) \\ = \frac{5}{4}t - \frac{5}{16} \sin 4t - \frac{5}{4}t \mathcal{U}(t-5) + \frac{5}{16} \sin 4(t-5) \mathcal{U}(t-5) + \frac{25}{4} \cos 4(t-5) \mathcal{U}(t-5).$$

- 72.** Recall from Section 3.8 that $mx'' = -kx + f(t)$. Now $m = W/g = 32/32 = 1$ slug, and $32 = 2k$ so that $k = 16$ lb/ft. Thus, the differential equation is $x'' + 16x = f(t)$. The initial conditions are $x(0) = 0$, $x'(0) = 0$. Also, since

$$f(t) = \begin{cases} \sin t, & 0 \leq t < 2\pi \\ 0, & t \geq 2\pi \end{cases}$$

and $\sin t = \sin(t-2\pi)$ we can write

$$f(t) = \sin t - \sin(t-2\pi) \mathcal{U}(t-2\pi).$$

The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{x\} + 16 \mathcal{L}\{x\} = \frac{1}{s^2+1} - \frac{1}{s^2+1} e^{-2\pi s}.$$

Solving for $\mathcal{L}\{x\}$ we obtain

$$\mathcal{L}\{x\} = \frac{1}{(s^2+16)(s^2+1)} - \frac{1}{(s^2+16)(s^2+1)} e^{-2\pi s} \\ = \frac{-1/15}{s^2+16} + \frac{1/15}{s^2+1} - \left[\frac{-1/15}{s^2+16} + \frac{1/15}{s^2+1} \right] e^{-2\pi s}.$$

Thus

$$\begin{aligned} x(t) &= -\frac{1}{60} \sin 4t + \frac{1}{15} \sin t + \frac{1}{60} \sin 4(t-2\pi) \mathcal{U}(t-2\pi) - \frac{1}{15} \sin(t-2\pi) \mathcal{U}(t-2\pi) \\ &= \begin{cases} -\frac{1}{60} \sin 4t + \frac{1}{15} \sin t, & 0 \leq t < 2\pi \\ 0, & t \geq 2\pi. \end{cases} \end{aligned}$$

73. The differential equation is

$$2.5 \frac{dq}{dt} + 12.5q = 5 \mathcal{U}(t-3).$$

The Laplace transform of this equation is

$$s \mathcal{L}\{q\} + 5 \mathcal{L}\{q\} = \frac{2}{s} e^{-3s}.$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \frac{2}{s(s+5)} e^{-3s} = \left(\frac{2}{5} \cdot \frac{1}{s} - \frac{2}{5} \cdot \frac{1}{s+5} \right) e^{-3s}.$$

Thus

$$q(t) = \frac{2}{5} \mathcal{U}(t-3) - \frac{2}{5} e^{-5(t-3)} \mathcal{U}(t-3).$$

74. The differential equation is

$$10 \frac{dq}{dt} + 10q = 30e^t - 30e^t \mathcal{U}(t-1.5).$$

The Laplace transform of this equation is

$$s \mathcal{L}\{q\} - q_0 + \mathcal{L}\{q\} = \frac{3}{s-1} - \frac{3e^{1.5}}{s-1.5} e^{-1.5s}.$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \left(q_0 - \frac{3}{2} \right) \cdot \frac{1}{s+1} + \frac{3}{2} \cdot \frac{1}{s-1} - 3e^{1.5} \left(\frac{-2/5}{s+1} + \frac{2/5}{s-1.5} \right) e^{-1.5s}.$$

Thus

$$q(t) = \left(q_0 - \frac{3}{2} \right) e^{-t} + \frac{3}{2} e^t + \frac{6}{5} e^{1.5} \left(e^{-(t-1.5)} - e^{1.5(t-1.5)} \right) \mathcal{U}(t-1.5).$$

75. (a) The differential equation is

$$\frac{di}{dt} + 10i = \sin t + \cos \left(t - \frac{3\pi}{2} \right) \mathcal{U} \left(t - \frac{3\pi}{2} \right), \quad i(0) = 0.$$

The Laplace transform of this equation is

$$s \mathcal{L}\{i\} + 10 \mathcal{L}\{i\} = \frac{1}{s^2+1} + \frac{se^{-3\pi s/2}}{s^2+1}.$$

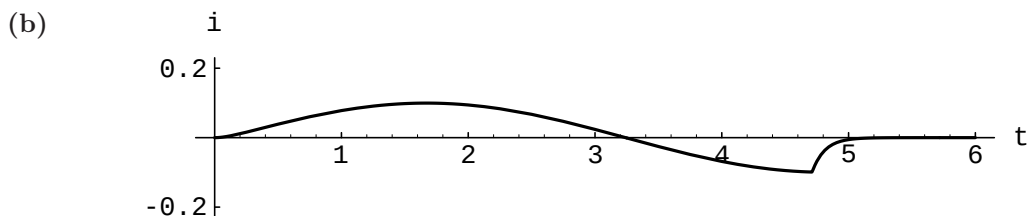
Solving for $\mathcal{L}\{i\}$ we obtain

$$\begin{aligned} \mathcal{L}\{i\} &= \frac{1}{(s^2+1)(s+10)} + \frac{s}{(s^2+1)(s+10)} e^{-3\pi s/2} \\ &= \frac{1}{101} \left(\frac{1}{s+10} - \frac{s}{s^2+1} + \frac{10}{s^2+1} \right) + \frac{1}{101} \left(\frac{-10}{s+10} + \frac{10s}{s^2+1} + \frac{1}{s^2+1} \right) e^{-3\pi s/2}. \end{aligned}$$

Thus

$$\begin{aligned} i(t) &= \frac{1}{101} (e^{-10t} - \cos t + 10 \sin t) \\ &\quad + \frac{1}{101} \left[-10e^{-10(t-3\pi/2)} + 10 \cos \left(t - \frac{3\pi}{2} \right) + \sin \left(t - \frac{3\pi}{2} \right) \right] \mathcal{U} \left(t - \frac{3\pi}{2} \right). \end{aligned}$$

4.3 Translation Theorems



The maximum value of $i(t)$ is approximately 0.1 at $t = 1.7$, the minimum is approximately -0.1 at 4.7 .

76. (a) The differential equation is

$$50 \frac{dq}{dt} + \frac{1}{0.01} q = E_0 [\mathcal{U}(t-1) - \mathcal{U}(t-3)], \quad q(0) = 0$$

or

$$50 \frac{dq}{dt} + 100q = E_0 [\mathcal{U}(t-1) - \mathcal{U}(t-3)], \quad q(0) = 0.$$

The Laplace transform of this equation is

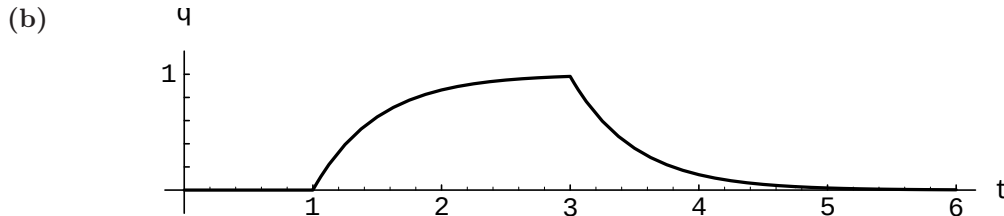
$$50s \mathcal{L}\{q\} + 100 \mathcal{L}\{q\} = E_0 \left(\frac{1}{s} e^{-s} - \frac{1}{s} e^{-3s} \right).$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \frac{E_0}{50} \left[\frac{e^{-s}}{s(s+2)} - \frac{e^{-3s}}{s(s+2)} \right] = \frac{E_0}{50} \left[\frac{1}{2} \left(\frac{1}{s} - \frac{1}{s+2} \right) e^{-s} - \frac{1}{2} \left(\frac{1}{s} - \frac{1}{s+2} \right) e^{-3s} \right].$$

Thus

$$q(t) = \frac{E_0}{100} \left[\left(1 - e^{-2(t-1)} \right) \mathcal{U}(t-1) - \left(1 - e^{-2(t-3)} \right) \mathcal{U}(t-3) \right].$$



The maximum value of $q(t)$ is approximately 1 at $t = 3$.

77. The differential equation is

$$EI \frac{d^4 y}{dx^4} = w_0 [1 - \mathcal{U}(x - L/2)].$$

Taking the Laplace transform of both sides and using $y(0) = y'(0) = 0$ we obtain

$$s^4 \mathcal{L}\{y\} - sy''(0) - y'''(0) = \frac{w_0}{EI} \frac{1}{s} (1 - e^{-Ls/2}).$$

Letting $y''(0) = c_1$ and $y'''(0) = c_2$ we have

$$\mathcal{L}\{y\} = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{w_0}{EI} \frac{1}{s^5} (1 - e^{-Ls/2})$$

so that

$$y(x) = \frac{1}{2} c_1 x^2 + \frac{1}{6} c_2 x^3 + \frac{1}{24} \frac{w_0}{EI} \left[x^4 - \left(x - \frac{L}{2} \right)^4 \mathcal{U}\left(x - \frac{L}{2} \right) \right].$$

To find c_1 and c_2 we compute

$$y''(x) = c_1 + c_2 x + \frac{1}{2} \frac{w_0}{EI} \left[x^2 - \left(x - \frac{L}{2} \right)^2 \mathcal{U}\left(x - \frac{L}{2} \right) \right]$$

and

$$y'''(x) = c_2 + \frac{w_0}{EI} \left[x - \left(x - \frac{L}{2} \right) \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

Then $y''(L) = y'''(L) = 0$ yields the system

$$\begin{aligned} c_1 + c_2 L + \frac{1}{2} \frac{w_0}{EI} \left[L^2 - \left(\frac{L}{2} \right)^2 \right] &= c_1 + c_2 L + \frac{3}{8} \frac{w_0 L^2}{EI} = 0 \\ c_2 + \frac{w_0}{EI} \left(\frac{L}{2} \right) &= c_2 + \frac{1}{2} \frac{w_0 L}{EI} = 0. \end{aligned}$$

Solving for c_1 and c_2 we obtain $c_1 = \frac{1}{8} w_0 L^2 / EI$ and $c_2 = -\frac{1}{2} w_0 L / EI$. Thus

$$y(x) = \frac{w_0}{EI} \left[\frac{1}{16} L^2 x^2 - \frac{1}{12} L x^3 + \frac{1}{24} x^4 - \frac{1}{24} \left(x - \frac{L}{2} \right)^4 \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

78. The differential equation is

$$EI \frac{d^4 y}{dx^4} = w_0 [\mathcal{U}(x - L/3) - \mathcal{U}(x - 2L/3)].$$

Taking the Laplace transform of both sides and using $y(0) = y'(0) = 0$ we obtain

$$s^4 \mathcal{L}\{y\} - s y''(0) - y'''(0) = \frac{w_0}{EI} \frac{1}{s} (e^{-Ls/3} - e^{-2Ls/3}).$$

Letting $y''(0) = c_1$ and $y'''(0) = c_2$ we have

$$\mathcal{L}\{y\} = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{w_0}{EI} \frac{1}{s^5} (e^{-Ls/3} - e^{-2Ls/3})$$

so that

$$y(x) = \frac{1}{2} c_1 x^2 + \frac{1}{6} c_2 x^3 + \frac{1}{24} \frac{w_0}{EI} \left[\left(x - \frac{L}{3} \right)^4 \mathcal{U} \left(x - \frac{L}{3} \right) - \left(x - \frac{2L}{3} \right)^4 \mathcal{U} \left(x - \frac{2L}{3} \right) \right].$$

To find c_1 and c_2 we compute

$$y''(x) = c_1 + c_2 x + \frac{1}{2} \frac{w_0}{EI} \left[\left(x - \frac{L}{3} \right)^2 \mathcal{U} \left(x - \frac{L}{3} \right) - \left(x - \frac{2L}{3} \right)^2 \mathcal{U} \left(x - \frac{2L}{3} \right) \right]$$

and

$$y'''(x) = c_2 + \frac{w_0}{EI} \left[\left(x - \frac{L}{3} \right) \mathcal{U} \left(x - \frac{L}{3} \right) - \left(x - \frac{2L}{3} \right) \mathcal{U} \left(x - \frac{2L}{3} \right) \right].$$

Then $y''(L) = y'''(L) = 0$ yields the system

$$\begin{aligned} c_1 + c_2 L + \frac{1}{2} \frac{w_0}{EI} \left[\left(\frac{2L}{3} \right)^2 - \left(\frac{L}{3} \right)^2 \right] &= c_1 + c_2 L + \frac{1}{6} \frac{w_0 L^2}{EI} = 0 \\ c_2 + \frac{w_0}{EI} \left[\frac{2L}{3} - \frac{L}{3} \right] &= c_2 + \frac{1}{3} \frac{w_0 L}{EI} = 0. \end{aligned}$$

Solving for c_1 and c_2 we obtain $c_1 = \frac{1}{6} w_0 L^2 / EI$ and $c_2 = -\frac{1}{3} w_0 L / EI$. Thus

$$y(x) = \frac{w_0}{EI} \left(\frac{1}{12} L^2 x^2 - \frac{1}{18} L x^3 + \frac{1}{24} \left[\left(x - \frac{L}{3} \right)^4 \mathcal{U} \left(x - \frac{L}{3} \right) - \left(x - \frac{2L}{3} \right)^4 \mathcal{U} \left(x - \frac{2L}{3} \right) \right] \right).$$

79. The differential equation is

$$EI \frac{d^4 y}{dx^4} = \frac{2w_0}{L} \left[\frac{L}{2} - x + \left(x - \frac{L}{2} \right) \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

4.3 Translation Theorems

Taking the Laplace transform of both sides and using $y(0) = y'(0) = 0$ we obtain

$$s^4 \mathcal{L}\{y\} - sy''(0) - y'''(0) = \frac{2w_0}{EIL} \left[\frac{L}{2s} - \frac{1}{s^2} + \frac{1}{s^2} e^{-Ls/2} \right].$$

Letting $y''(0) = c_1$ and $y'''(0) = c_2$ we have

$$\mathcal{L}\{y\} = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{2w_0}{EIL} \left[\frac{L}{2s^5} - \frac{1}{s^6} + \frac{1}{s^6} e^{-Ls/2} \right]$$

so that

$$\begin{aligned} y(x) &= \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3 + \frac{2w_0}{EIL} \left[\frac{L}{48}x^4 - \frac{1}{120}x^5 + \frac{1}{120} \left(x - \frac{L}{2} \right)^5 \mathcal{U} \left(x - \frac{L}{2} \right) \right] \\ &= \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3 + \frac{w_0}{60EIL} \left[\frac{5L}{2}x^4 - x^5 + \left(x - \frac{L}{2} \right)^5 \mathcal{U} \left(x - \frac{L}{2} \right) \right]. \end{aligned}$$

To find c_1 and c_2 we compute

$$y''(x) = c_1 + c_2x + \frac{w_0}{60EIL} \left[30Lx^2 - 20x^3 + 20 \left(x - \frac{L}{2} \right)^3 \mathcal{U} \left(x - \frac{L}{2} \right) \right]$$

and

$$y'''(x) = c_2 + \frac{w_0}{60EIL} \left[60Lx - 60x^2 + 60 \left(x - \frac{L}{2} \right)^2 \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

Then $y''(L) = y'''(L) = 0$ yields the system

$$\begin{aligned} c_1 + c_2L + \frac{w_0}{60EIL} \left[30L^3 - 20L^3 + \frac{5}{2}L^3 \right] &= c_1 + c_2L + \frac{5w_0L^2}{24EI} = 0 \\ c_2 + \frac{w_0}{60EIL} [60L^2 - 60L^2 + 15L^2] &= c_2 + \frac{w_0L}{4EI} = 0. \end{aligned}$$

Solving for c_1 and c_2 we obtain $c_1 = w_0L^2/24EI$ and $c_2 = -w_0L/4EI$. Thus

$$y(x) = \frac{w_0L^2}{48EI}x^2 - \frac{w_0L}{24EI}x^3 + \frac{w_0}{60EIL} \left[\frac{5L}{2}x^4 - x^5 + \left(x - \frac{L}{2} \right)^5 \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

80. The differential equation is

$$EI \frac{d^4y}{dx^4} = w_0[1 - \mathcal{U}(x - L/2)].$$

Taking the Laplace transform of both sides and using $y(0) = y'(0) = 0$ we obtain

$$s^4 \mathcal{L}\{y\} - sy''(0) - y'''(0) = \frac{w_0}{EI} \frac{1}{s} (1 - e^{-Ls/2}).$$

Letting $y''(0) = c_1$ and $y'''(0) = c_2$ we have

$$\mathcal{L}\{y\} = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{w_0}{EI} \frac{1}{s^5} (1 - e^{-Ls/2})$$

so that

$$y(x) = \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3 + \frac{1}{24} \frac{w_0}{EI} \left[x^4 - \left(x - \frac{L}{2} \right)^4 \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

To find c_1 and c_2 we compute

$$y''(x) = c_1 + c_2x + \frac{1}{2} \frac{w_0}{EI} \left[x^2 - \left(x - \frac{L}{2} \right)^2 \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

Then $y(L) = y''(L) = 0$ yields the system

$$\begin{aligned}\frac{1}{2}c_1L^2 + \frac{1}{6}c_2L^3 + \frac{1}{24}\frac{w_0}{EI}\left[L^4 - \left(\frac{L}{2}\right)^4\right] &= \frac{1}{2}c_1L^2 + \frac{1}{6}c_2L^3 + \frac{5w_0}{128EI}L^4 = 0 \\ c_1 + c_2L + \frac{1}{2}\frac{w_0}{EI}\left[L^2 - \left(\frac{L}{2}\right)^2\right] &= c_1 + c_2L + \frac{3w_0}{8EI}L^2 = 0.\end{aligned}$$

Solving for c_1 and c_2 we obtain $c_1 = \frac{9}{128}w_0L^2/EI$ and $c_2 = -\frac{57}{128}w_0L/EI$. Thus

$$y(x) = \frac{w_0}{EI}\left[\frac{9}{256}L^2x^2 - \frac{19}{256}Lx^3 + \frac{1}{24}x^4 - \frac{1}{24}\left(x - \frac{L}{2}\right)^4\mathcal{U}\left(x - \frac{L}{2}\right)\right].$$

81. (a) The temperature T of the cake inside the oven is modeled by

$$\frac{dT}{dt} = k(T - T_m)$$

where T_m is the ambient temperature of the oven. For $0 \leq t \leq 4$, we have

$$T_m = 70 + \frac{300 - 70}{4 - 0}t = 70 + 57.5t.$$

Hence for $t \geq 0$,

$$T_m = \begin{cases} 70 + 57.5t, & 0 \leq t < 4 \\ 300, & t \geq 4. \end{cases}$$

In terms of the unit step function,

$$T_m = (70 + 57.5t)[1 - \mathcal{U}(t - 4)] + 300\mathcal{U}(t - 4) = 70 + 57.5t + (230 - 57.5t)\mathcal{U}(t - 4).$$

The initial-value problem is then

$$\frac{dT}{dt} = k[T - 70 - 57.5t - (230 - 57.5t)\mathcal{U}(t - 4)], \quad T(0) = 70.$$

- (b) Let $t(s) = \mathcal{L}\{T(t)\}$. Transforming the equation, using $230 - 57.5t = -57.5(t - 4)$ and Theorem 4.7, gives

$$st(s) - 70 = k\left(t(s) - \frac{70}{s} - \frac{57.5}{s^2} + \frac{57.5}{s^2}e^{-4s}\right)$$

or

$$t(s) = \frac{70}{s - k} - \frac{70k}{s(s - k)} - \frac{57.5k}{s^2(s - k)} + \frac{57.5k}{s^2(s - k)}e^{-4s}.$$

After using partial functions, the inverse transform is then

$$T(t) = 70 + 57.5\left(\frac{1}{k} + t - \frac{1}{k}e^{kt}\right) - 57.5\left(\frac{1}{k} + t - 4 - \frac{1}{k}e^{k(t-4)}\right)\mathcal{U}(t - 4).$$

Of course, the obvious question is: What is k ? If the cake is supposed to bake for, say, 20 minutes, then $T(20) = 300$. That is,

$$300 = 70 + 57.5\left(\frac{1}{k} + 20 - \frac{1}{k}e^{20k}\right) - 57.5\left(\frac{1}{k} + 16 - \frac{1}{k}e^{16k}\right).$$

But this equation has no physically meaningful solution. This should be no surprise since the model predicts the asymptotic behavior $T(t) \rightarrow 300$ as t increases. Using $T(20) = 299$ instead, we find, with the help of a CAS, that $k \approx -0.3$.

82. In order to apply Theorem 4.7 we need the function to have the form $f(t - a)\mathcal{U}(t - a)$. To accomplish this rewrite the functions given in the forms shown below.

4.3 Translation Theorems

(a) $2t + 1 = 2(t - 1 + 1) + 1 = 2(t - 1) + 3$

(b) $e^t = e^{t-5+5} = e^5 e^{t-5}$

(c) $\cos t = -\cos(t - \pi)$

(d) $t^2 - 3t = (t - 2)^2 + (t - 2) - 2$

83. (a) From Theorem 4.6 we have $\mathcal{L}\{te^{kti}\} = 1/(s - ki)^2$. Then, using Euler's formula,

$$\begin{aligned}\mathcal{L}\{te^{kti}\} &= \mathcal{L}\{t \cos kt + it \sin kt\} = \mathcal{L}\{t \cos kt\} + i \mathcal{L}\{t \sin kt\} \\ &= \frac{1}{(s - ki)^2} = \frac{(s + ki)^2}{(s^2 + k^2)^2} = \frac{s^2 - k^2}{(s^2 + k^2)^2} + i \frac{2ks}{(s^2 + k^2)^2}.\end{aligned}$$

Equating real and imaginary parts we have

$$\mathcal{L}\{t \cos kt\} = \frac{s^2 - k^2}{(s^2 + k^2)^2} \quad \text{and} \quad \mathcal{L}\{t \sin kt\} = \frac{2ks}{(s^2 + k^2)^2}.$$

(b) The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{x\} + \omega^2 \mathcal{L}\{x\} = \frac{s}{s^2 + \omega^2}.$$

Solving for $\mathcal{L}\{x\}$ we obtain $\mathcal{L}\{x\} = s/(s^2 + \omega^2)^2$. Thus $x = (1/2\omega)t \sin \omega t$.

EXERCISES 4.4

Additional Operational Properties

1. $\mathcal{L}\{te^{-10t}\} = -\frac{d}{ds} \left(\frac{1}{s + 10} \right) = \frac{1}{(s + 10)^2}$

2. $\mathcal{L}\{t^3 e^t\} = (-1)^3 \frac{d^3}{ds^3} \left(\frac{1}{s - 1} \right) = \frac{6}{(s - 1)^4}$

3. $\mathcal{L}\{t \cos 2t\} = -\frac{d}{ds} \left(\frac{s}{s^2 + 4} \right) = \frac{s^2 - 4}{(s^2 + 4)^2}$

4. $\mathcal{L}\{t \sinh 3t\} = -\frac{d}{ds} \left(\frac{3}{s^2 - 9} \right) = \frac{6s}{(s^2 - 9)^2}$

5. $\mathcal{L}\{t^2 \sinh t\} = \frac{d^2}{ds^2} \left(\frac{1}{s^2 - 1} \right) = \frac{6s^2 + 2}{(s^2 - 1)^3}$

6. $\mathcal{L}\{t^2 \cos t\} = \frac{d^2}{ds^2} \left(\frac{s}{s^2 + 1} \right) = \frac{d}{ds} \left(\frac{1 - s^2}{(s^2 + 1)^2} \right) = \frac{2s(s^2 - 3)}{(s^2 + 1)^3}$

7. $\mathcal{L}\{te^{2t} \sin 6t\} = -\frac{d}{ds} \left(\frac{6}{(s - 2)^2 + 36} \right) = \frac{12(s - 2)}{[(s - 2)^2 + 36]^2}$

8. $\mathcal{L}\{te^{-3t} \cos 3t\} = -\frac{d}{ds} \left(\frac{s + 3}{(s + 3)^2 + 9} \right) = \frac{(s + 3)^2 - 9}{[(s + 3)^2 + 9]^2}$

9. The Laplace transform of the differential equation is

$$s \mathcal{L}\{y\} + \mathcal{L}\{y\} = \frac{2s}{(s^2 + 1)^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2s}{(s + 1)(s^2 + 1)^2} = -\frac{1}{2} \frac{1}{s + 1} - \frac{1}{2} \frac{1}{s^2 + 1} + \frac{1}{2} \frac{s}{s^2 + 1} + \frac{1}{(s^2 + 1)^2} + \frac{s}{(s^2 + 1)^2}.$$

Thus

$$\begin{aligned} y(t) &= -\frac{1}{2}e^{-t} - \frac{1}{2}\sin t + \frac{1}{2}\cos t + \frac{1}{2}(\sin t - t\cos t) + \frac{1}{2}t\sin t \\ &= -\frac{1}{2}e^{-t} + \frac{1}{2}\cos t - \frac{1}{2}t\cos t + \frac{1}{2}t\sin t. \end{aligned}$$

10. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - \mathcal{L}\{y\} = \frac{2(s-1)}{((s-1)^2+1)^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2}{((s-1)^2+1)^2}.$$

Thus

$$y = e^t \sin t - te^t \cos t.$$

11. The Laplace transform of the differential equation is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) + 9\mathcal{L}\{y\} = \frac{s}{s^2+9}.$$

Letting $y(0) = 2$ and $y'(0) = 5$ and solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2s^3 + 5s^2 + 19s - 45}{(s^2+9)^2} = \frac{2s}{s^2+9} + \frac{5}{s^2+9} + \frac{s}{(s^2+9)^2}.$$

Thus

$$y = 2\cos 3t + \frac{5}{3}\sin 3t + \frac{1}{6}t\sin 3t.$$

12. The Laplace transform of the differential equation is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) + \mathcal{L}\{y\} = \frac{1}{s^2+1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s^3 - s^2 + s}{(s^2+1)^2} = \frac{s}{s^2+1} - \frac{1}{s^2+1} + \frac{1}{(s^2+1)^2}.$$

Thus

$$y = \cos t - \sin t + \left(\frac{1}{2}\sin t - \frac{1}{2}t\cos t\right) = \cos t - \frac{1}{2}\sin t - \frac{1}{2}t\cos t.$$

13. The Laplace transform of the differential equation is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) + 16\mathcal{L}\{y\} = \mathcal{L}\{\cos 4t - \cos 4t\mathcal{U}(t-\pi)\}$$

or by (16) of Section 4.3 in the text,

$$\begin{aligned} (s^2+16)\mathcal{L}\{y\} &= 1 + \frac{s}{s^2+16} - e^{-\pi s}\mathcal{L}\{\cos 4(t+\pi)\} \\ &= 1 + \frac{s}{s^2+16} - e^{-\pi s}\mathcal{L}\{\cos 4t\} = 1 + \frac{s}{s^2+16} - \frac{s}{s^2+16}e^{-\pi s}. \end{aligned}$$

Thus

$$\mathcal{L}\{y\} = \frac{1}{s^2+16} + \frac{s}{(s^2+16)^2} - \frac{s}{(s^2+16)^2}e^{-\pi s}$$

and

$$y = \frac{1}{4}\sin 4t + \frac{1}{8}t\sin 4t - \frac{1}{8}(t-\pi)\sin 4(t-\pi)\mathcal{U}(t-\pi).$$

4.4 Additional Operational Properties

14. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + \mathcal{L}\{y\} = \mathcal{L}\left\{1 - \mathcal{U}\left(t - \frac{\pi}{2}\right) + \sin t \mathcal{U}\left(t - \frac{\pi}{2}\right)\right\}$$

or

$$\begin{aligned}(s^2 + 1) \mathcal{L}\{y\} &= s + \frac{1}{s} - \frac{1}{s} e^{-\pi s/2} + e^{-\pi s/2} \mathcal{L}\left\{\sin\left(t + \frac{\pi}{2}\right)\right\} \\ &= s + \frac{1}{s} - \frac{1}{s} e^{-\pi s/2} + e^{-\pi s/2} \mathcal{L}\{\cos t\} \\ &= s + \frac{1}{s} - \frac{1}{s} e^{-\pi s/2} + \frac{s}{s^2 + 1} e^{-\pi s/2}.\end{aligned}$$

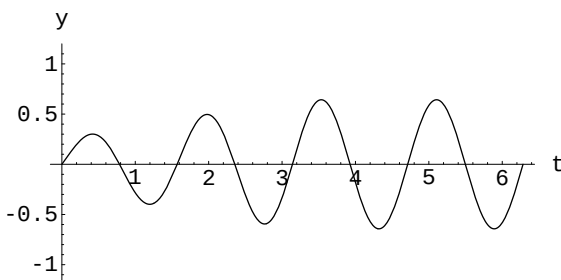
Thus

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{s}{s^2 + 1} + \frac{1}{s(s^2 + 1)} - \frac{1}{s(s^2 + 1)} e^{-\pi s/2} + \frac{s}{(s^2 + 1)^2} e^{-\pi s/2} \\ &= \frac{s}{s^2 + 1} + \frac{1}{s} - \frac{s}{s^2 + 1} - \left(\frac{1}{s} - \frac{s}{s^2 + 1}\right) e^{-\pi s/2} + \frac{s}{(s^2 + 1)^2} e^{-\pi s/2} \\ &= \frac{1}{s} - \left(\frac{1}{s} - \frac{s}{s^2 + 1}\right) e^{-\pi s/2} + \frac{s}{(s^2 + 1)^2} e^{-\pi s/2}\end{aligned}$$

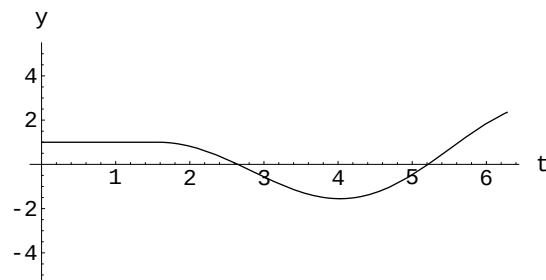
and

$$\begin{aligned}y &= 1 - \left[1 - \cos\left(t - \frac{\pi}{2}\right)\right] \mathcal{U}\left(t - \frac{\pi}{2}\right) + \frac{1}{2} \left(t - \frac{\pi}{2}\right) \sin\left(t - \frac{\pi}{2}\right) \mathcal{U}\left(t - \frac{\pi}{2}\right) \\ &= 1 - (1 - \sin t) \mathcal{U}\left(t - \frac{\pi}{2}\right) - \frac{1}{2} \left(t - \frac{\pi}{2}\right) \cos t \mathcal{U}\left(t - \frac{\pi}{2}\right).\end{aligned}$$

15.



16.



17. From (7) of Section 4.2 in the text along with Theorem 4.8,

$$\mathcal{L}\{ty''\} = -\frac{d}{ds} \mathcal{L}\{y''\} = -\frac{d}{ds}[s^2 Y(s) - sy(0) - y'(0)] = -s^2 \frac{dY}{ds} - 2sY + y(0),$$

so that the transform of the given second-order differential equation is the linear first-order differential equation in $Y(s)$:

$$s^2 Y' + 3sY = -\frac{4}{s^3} \quad \text{or} \quad Y' + \frac{3}{s} Y = -\frac{4}{s^5}.$$

The solution of the latter equation is $Y(s) = 4/s^4 + c/s^3$, so

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \frac{2}{3} t^3 + \frac{c}{2} t^2.$$

18. From Theorem 4.8 in the text

$$\mathcal{L}\{ty'\} = -\frac{d}{ds} \mathcal{L}\{y'\} = -\frac{d}{ds}[sY(s) - y(0)] = -s \frac{dY}{ds} - Y$$

so that the transform of the given second-order differential equation is the linear first-order differential equation in $Y(s)$:

$$Y' + \left(\frac{3}{s} - 2s\right) Y = -\frac{10}{s}.$$

Using the integrating factor $s^3 e^{-s^2}$, the last equation yields

$$Y(s) = \frac{5}{s^3} + \frac{c}{s^3} e^{s^2}.$$

But if $Y(s)$ is the Laplace transform of a piecewise-continuous function of exponential order, we must have, in view of Theorem 4.5, $\lim_{s \rightarrow \infty} Y(s) = 0$. In order to obtain this condition we require $c = 0$. Hence

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{5}{s^3} \right\} = \frac{5}{2} t^2.$$

$$19. \quad \mathcal{L}\{1 * t^3\} = \frac{1}{s} \frac{3!}{s^4} = \frac{6}{s^5}$$

$$20. \quad \mathcal{L}\{t^2 * te^t\} = \frac{2}{s^3(s-1)^2}$$

$$21. \quad \mathcal{L}\{e^{-t} * e^t \cos t\} = \frac{s-1}{(s+1)[(s-1)^2+1]}$$

$$22. \quad \mathcal{L}\{e^{2t} * \sin t\} = \frac{1}{(s-2)(s^2+1)}$$

$$23. \quad \mathcal{L}\left\{\int_0^t e^\tau d\tau\right\} = \frac{1}{s} \mathcal{L}\{e^t\} = \frac{1}{s(s-1)}$$

$$24. \quad \mathcal{L}\left\{\int_0^t \cos \tau d\tau\right\} = \frac{1}{s} \mathcal{L}\{\cos t\} = \frac{s}{s(s^2+1)} = \frac{1}{s^2+1}$$

$$25. \quad \mathcal{L}\left\{\int_0^t e^{-\tau} \cos \tau d\tau\right\} = \frac{1}{s} \mathcal{L}\{e^{-t} \cos t\} = \frac{1}{s} \frac{s+1}{(s+1)^2+1} = \frac{s+1}{s(s^2+2s+2)}$$

$$26. \quad \mathcal{L}\left\{\int_0^t \tau \sin \tau d\tau\right\} = \frac{1}{s} \mathcal{L}\{t \sin t\} = \frac{1}{s} \left(-\frac{d}{ds} \frac{1}{s^2+1}\right) = -\frac{1}{s} \frac{-2s}{(s^2+1)^2} = \frac{2}{(s^2+1)^2}$$

$$27. \quad \mathcal{L}\left\{\int_0^t \tau e^{t-\tau} d\tau\right\} = \mathcal{L}\{t\} \mathcal{L}\{e^t\} = \frac{1}{s^2(s-1)}$$

$$28. \quad \mathcal{L}\left\{\int_0^t \sin \tau \cos(t-\tau) d\tau\right\} = \mathcal{L}\{\sin t\} \mathcal{L}\{\cos t\} = \frac{s}{(s^2+1)^2}$$

$$29. \quad \mathcal{L}\left\{t \int_0^t \sin \tau d\tau\right\} = -\frac{d}{ds} \mathcal{L}\left\{\int_0^t \sin \tau d\tau\right\} = -\frac{d}{ds} \left(\frac{1}{s} \frac{1}{s^2+1}\right) = \frac{3s^2+1}{s^2(s^2+1)^2}$$

$$30. \quad \mathcal{L}\left\{t \int_0^t \tau e^{-\tau} d\tau\right\} = -\frac{d}{ds} \mathcal{L}\left\{\int_0^t \tau e^{-\tau} d\tau\right\} = -\frac{d}{ds} \left(\frac{1}{s} \frac{1}{(s+1)^2}\right) = \frac{3s+1}{s^2(s+1)^3}$$

$$31. \quad \mathcal{L}^{-1}\left\{\frac{1}{s(s-1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/(s-1)}{s}\right\} = \int_0^t e^\tau d\tau = e^t - 1$$

$$32. \quad \mathcal{L}^{-1}\left\{\frac{1}{s^2(s-1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/s(s-1)}{s}\right\} = \int_0^t (e^\tau - 1) d\tau = e^t - t - 1$$

$$33. \quad \mathcal{L}^{-1}\left\{\frac{1}{s^3(s-1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/s^2(s-1)}{s}\right\} = \int_0^t (e^\tau - \tau - 1) d\tau = e^t - \frac{1}{2}t^2 - t - 1$$

$$34. \quad \text{Using } \mathcal{L}^{-1}\left\{\frac{1}{(s-a)^2}\right\} = te^{at}, (8) \text{ in the text gives}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s-a)^2}\right\} = \int_0^t \tau e^{a\tau} d\tau = \frac{1}{a^2}(ate^{at} - e^{at} + 1).$$

$$35. \quad (\text{a}) \quad \text{The result in (4) in the text is } \mathcal{L}^{-1}\{F(s)G(s)\} = f * g, \text{ so identify}$$

$$F(s) = \frac{2k^3}{(s^2+k^2)^2} \quad \text{and} \quad G(s) = \frac{4s}{s^2+k^2}.$$

4.4 Additional Operational Properties

Then

$$f(t) = \sin kt - kt \cos kt \quad \text{and} \quad g(t) = 4 \cos kt$$

so

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{8k^3 s}{(s^2 + k^2)^3} \right\} &= \mathcal{L}^{-1} \{F(s)G(s)\} = f * g = 4 \int_0^t f(\tau)g(t - \tau)d\tau \\ &= 4 \int_0^t (\sin k\tau - k\tau \cos k\tau) \cos k(t - \tau)d\tau. \end{aligned}$$

Using a CAS to evaluate the integral we get

$$\mathcal{L}^{-1} \left\{ \frac{8k^3 s}{(s^2 + k^2)^3} \right\} = t \sin kt - kt^2 \cos kt.$$

(b) Observe from part (a) that

$$\mathcal{L}\{t(\sin kt - kt \cos kt)\} = \frac{8k^3 s}{(s^2 + k^2)^3},$$

and from Theorem 4.8 that $\mathcal{L}\{tf(t)\} = -F'(s)$. We saw in (5) in the text that

$$\mathcal{L}\{\sin kt - kt \cos kt\} = 2k^3/(s^2 + k^2)^2,$$

so

$$\mathcal{L}\{t(\sin kt - kt \cos kt)\} = -\frac{d}{ds} \frac{2k^3}{(s^2 + k^2)^2} = \frac{8k^3 s}{(s^2 + k^2)^3}.$$

36. The Laplace transform of the differential equation is

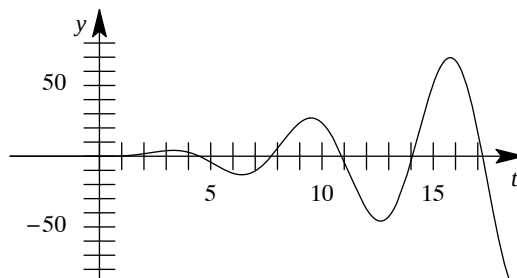
$$s^2 \mathcal{L}\{y\} + \mathcal{L}\{y\} = \frac{1}{(s^2 + 1)} + \frac{2s}{(s^2 + 1)^2}.$$

Thus

$$\mathcal{L}\{y\} = \frac{1}{(s^2 + 1)^2} + \frac{2s}{(s^2 + 1)^3}$$

and, using Problem 35 with $k = 1$,

$$y = \frac{1}{2}(\sin t - t \cos t) + \frac{1}{4}(t \sin t - t^2 \cos t).$$



37. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} + \mathcal{L}\{t\} \mathcal{L}\{f\} = \mathcal{L}\{t\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain $\mathcal{L}\{f\} = \frac{1}{s^2 + 1}$. Thus, $f(t) = \sin t$.

38. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} = \mathcal{L}\{2t\} - 4 \mathcal{L}\{\sin t\} \mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{2s^2 + 2}{s^2(s^2 + 5)} = \frac{2}{5} \frac{1}{s^2} + \frac{8}{5\sqrt{5}} \frac{\sqrt{5}}{s^2 + 5}.$$

Thus

$$f(t) = \frac{2}{5}t + \frac{8}{5\sqrt{5}} \sin \sqrt{5}t.$$

39. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} = \mathcal{L}\{te^t\} + \mathcal{L}\{t\} \mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{s^2}{(s-1)^3(s+1)} = \frac{1}{8} \frac{1}{s-1} + \frac{3}{4} \frac{1}{(s-1)^2} + \frac{1}{4} \frac{2}{(s-1)^3} - \frac{1}{8} \frac{1}{s+1}.$$

Thus

$$f(t) = \frac{1}{8}e^t + \frac{3}{4}te^t + \frac{1}{4}t^2e^t - \frac{1}{8}e^{-t}$$

40. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} + 2\mathcal{L}\{\cos t\}\mathcal{L}\{f\} = 4\mathcal{L}\{e^{-t}\} + \mathcal{L}\{\sin t\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{4s^2 + s + 5}{(s+1)^3} = \frac{4}{s+1} - \frac{7}{(s+1)^2} + 4\frac{2}{(s+1)^3}.$$

Thus

$$f(t) = 4e^{-t} - 7te^{-t} + 4t^2e^{-t}.$$

41. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} + \mathcal{L}\{1\}\mathcal{L}\{f\} = \mathcal{L}\{1\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain $\mathcal{L}\{f\} = \frac{1}{s+1}$. Thus, $f(t) = e^{-t}$.

42. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} = \mathcal{L}\{\cos t\} + \mathcal{L}\{e^{-t}\}\mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{s}{s^2+1} + \frac{1}{s^2+1}.$$

Thus

$$f(t) = \cos t + \sin t.$$

43. The Laplace transform of the given equation is

$$\begin{aligned}\mathcal{L}\{f\} &= \mathcal{L}\{1\} + \mathcal{L}\{t\} - \mathcal{L}\left\{\frac{8}{3}\int_0^t (t-\tau)^3 f(\tau) d\tau\right\} \\ &= \frac{1}{s} + \frac{1}{s^2} + \frac{8}{3}\mathcal{L}\{t^3\}\mathcal{L}\{f\} = \frac{1}{s} + \frac{1}{s^2} + \frac{16}{s^4}\mathcal{L}\{f\}.\end{aligned}$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{s^2(s+1)}{s^4-16} = \frac{1}{8} \frac{1}{s+2} + \frac{3}{8} \frac{1}{s-2} + \frac{1}{4} \frac{2}{s^2+4} + \frac{1}{2} \frac{s}{s^2+4}.$$

Thus

$$f(t) = \frac{1}{8}e^{-2t} + \frac{3}{8}e^{2t} + \frac{1}{4}\sin 2t + \frac{1}{2}\cos 2t.$$

44. The Laplace transform of the given equation is

$$\mathcal{L}\{t\} - 2\mathcal{L}\{f\} = \mathcal{L}\{e^t - e^{-t}\}\mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{s^2-1}{2s^4} = \frac{1}{2} \frac{1}{s^2} - \frac{1}{12} \frac{3!}{s^4}.$$

4.4 Additional Operational Properties

Thus

$$f(t) = \frac{1}{2}t - \frac{1}{12}t^3.$$

45. The Laplace transform of the given equation is

$$s \mathcal{L}\{y\} - y(0) = \mathcal{L}\{1\} - \mathcal{L}\{\sin t\} - \mathcal{L}\{1\} \mathcal{L}\{y\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s^2 - s + 1}{(s^2 + 1)^2} = \frac{1}{s^2 + 1} - \frac{1}{2} \frac{2s}{(s^2 + 1)^2}.$$

Thus

$$y = \sin t - \frac{1}{2}t \sin t.$$

46. The Laplace transform of the given equation is

$$s \mathcal{L}\{y\} - y(0) + 6 \mathcal{L}\{y\} + 9 \mathcal{L}\{1\} \mathcal{L}\{y\} = \mathcal{L}\{1\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain $\mathcal{L}\{y\} = \frac{1}{(s+3)^2}$. Thus, $y = te^{-3t}$.

47. The differential equation is

$$0.1 \frac{di}{dt} + 3i + \frac{1}{0.05} \int_0^t i(\tau) d\tau = 100 [\mathcal{U}(t-1) - \mathcal{U}(t-2)]$$

or

$$\frac{di}{dt} + 30i + 200 \int_0^t i(\tau) d\tau = 1000 [\mathcal{U}(t-1) - \mathcal{U}(t-2)],$$

where $i(0) = 0$. The Laplace transform of the differential equation is

$$s \mathcal{L}\{i\} - y(0) + 30 \mathcal{L}\{i\} + \frac{200}{s} \mathcal{L}\{i\} = \frac{1000}{s} (e^{-s} - e^{-2s}).$$

Solving for $\mathcal{L}\{i\}$ we obtain

$$\mathcal{L}\{i\} = \frac{1000e^{-s} - 1000e^{-2s}}{s^2 + 30s + 200} = \left(\frac{100}{s+10} - \frac{100}{s+20} \right) (e^{-s} - e^{-2s}).$$

Thus

$$i(t) = 100(e^{-10(t-1)} - e^{-20(t-1)}) \mathcal{U}(t-1) - 100(e^{-10(t-2)} - e^{-20(t-2)}) \mathcal{U}(t-2).$$

48. The differential equation is

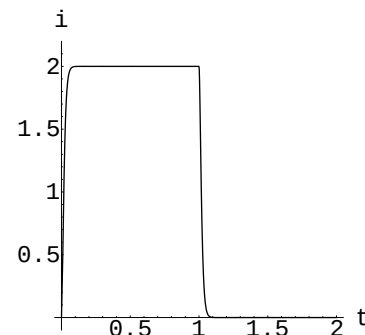
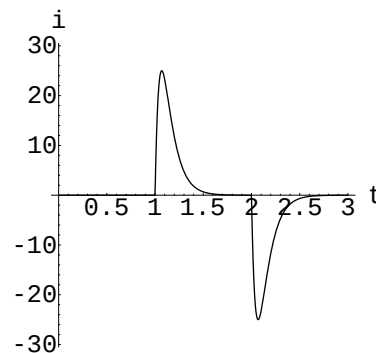
$$0.005 \frac{di}{dt} + i + \frac{1}{0.02} \int_0^t i(\tau) d\tau = 100 [t - (t-1) \mathcal{U}(t-1)]$$

or

$$\frac{di}{dt} + 200i + 10,000 \int_0^t i(\tau) d\tau = 20,000 [t - (t-1) \mathcal{U}(t-1)],$$

where $i(0) = 0$. The Laplace transform of the differential equation is

$$s \mathcal{L}\{i\} + 200 \mathcal{L}\{i\} + \frac{10,000}{s} \mathcal{L}\{i\} = 20,000 \left(\frac{1}{s^2} - \frac{1}{s^2} e^{-s} \right).$$



Solving for $\mathcal{L}\{i\}$ we obtain

$$\mathcal{L}\{i\} = \frac{20,000}{s(s+100)^2}(1-e^{-s}) = \left[\frac{2}{s} - \frac{2}{s+100} - \frac{200}{(s+100)^2} \right] (1-e^{-s}).$$

Thus

$$i(t) = 2 - 2e^{-100t} - 200te^{-100t} - 2\mathcal{U}(t-1) + 2e^{-100(t-1)}\mathcal{U}(t-1) + 200(t-1)e^{-100(t-1)}\mathcal{U}(t-1).$$

$$49. \quad \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2as}} \left[\int_0^a e^{-st} dt - \int_a^{2a} e^{-st} dt \right] = \frac{(1-e^{-as})^2}{s(1-e^{-2as})} = \frac{1-e^{-as}}{s(1+e^{-as})}$$

$$50. \quad \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2as}} \int_0^a e^{-st} dt = \frac{1}{s(1+e^{-as})}$$

51. Using integration by parts,

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-bs}} \int_0^b \frac{a}{b} te^{-st} dt = \frac{a}{s} \left(\frac{1}{bs} - \frac{1}{e^{bs}-1} \right).$$

$$52. \quad \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2s}} \left[\int_0^1 te^{-st} dt + \int_1^2 (2-t)e^{-st} dt \right] = \frac{1-e^{-s}}{s^2(1-e^{-2s})}$$

$$53. \quad \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-\pi s}} \int_0^\pi e^{-st} \sin t dt = \frac{1}{s^2+1} \cdot \frac{e^{\pi s/2} + e^{-\pi s/2}}{e^{\pi s/2} - e^{-\pi s/2}} = \frac{1}{s^2+1} \coth \frac{\pi s}{2}$$

$$54. \quad \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2\pi s}} \int_0^\pi e^{-st} \sin t dt = \frac{1}{s^2+1} \cdot \frac{1}{1-e^{-\pi s}}$$

55. The differential equation is $L di/dt + Ri = E(t)$, where $i(0) = 0$. The Laplace transform of the equation is

$$Ls \mathcal{L}\{i\} + R \mathcal{L}\{i\} = \mathcal{L}\{E(t)\}.$$

From Problem 49 we have $\mathcal{L}\{E(t)\} = (1-e^{-s})/s(1+e^{-s})$. Thus

$$(Ls + R) \mathcal{L}\{i\} = \frac{1-e^{-s}}{s(1+e^{-s})}$$

and

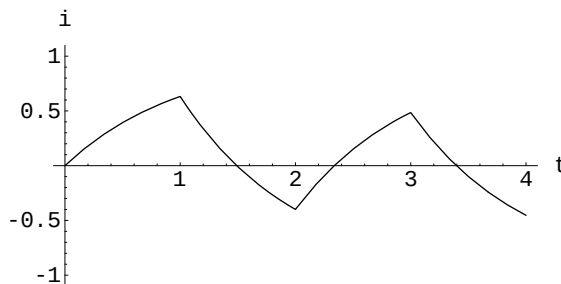
$$\begin{aligned} \mathcal{L}\{i\} &= \frac{1}{L} \frac{1-e^{-s}}{s(s+R/L)(1+e^{-s})} = \frac{1}{L} \frac{1-e^{-s}}{s(s+R/L)} \frac{1}{1+e^{-s}} \\ &= \frac{1}{R} \left(\frac{1}{s} - \frac{1}{s+R/L} \right) (1-e^{-s})(1-e^{-s}+e^{-2s}-e^{-3s}+e^{-4s}-\dots) \\ &= \frac{1}{R} \left(\frac{1}{s} - \frac{1}{s+R/L} \right) (1-2e^{-s}+2e^{-2s}-2e^{-3s}+2e^{-4s}-\dots). \end{aligned}$$

Therefore,

$$\begin{aligned} i(t) &= \frac{1}{R} \left(1 - e^{-Rt/L} \right) - \frac{2}{R} \left(1 - e^{-R(t-1)/L} \right) \mathcal{U}(t-1) \\ &\quad + \frac{2}{R} \left(1 - e^{-R(t-2)/L} \right) \mathcal{U}(t-2) - \frac{2}{R} \left(1 - e^{-R(t-3)/L} \right) \mathcal{U}(t-3) + \dots \\ &= \frac{1}{R} \left(1 - e^{-Rt/L} \right) + \frac{2}{R} \sum_{n=1}^{\infty} \left(1 - e^{-R(t-n)/L} \right) \mathcal{U}(t-n). \end{aligned}$$

4.4 Additional Operational Properties

The graph of $i(t)$ with $L = 1$ and $R = 1$ is shown below.



56. The differential equation is $L di/dt + Ri = E(t)$, where $i(0) = 0$. The Laplace transform of the equation is

$$Ls \mathcal{L}\{i\} + R \mathcal{L}\{i\} = \mathcal{L}\{E(t)\}.$$

From Problem 51 we have

$$\mathcal{L}\{E(t)\} = \frac{1}{s} \left(\frac{1}{s} - \frac{1}{e^s - 1} \right) = \frac{1}{s^2} - \frac{1}{s} \frac{1}{e^s - 1}.$$

Thus

$$(Ls + R) \mathcal{L}\{i\} = \frac{1}{s^2} - \frac{1}{s} \frac{1}{e^s - 1}$$

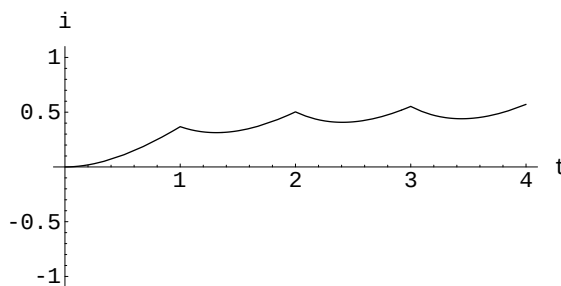
and

$$\begin{aligned} \mathcal{L}\{i\} &= \frac{1}{L} \frac{1}{s^2(s + R/L)} - \frac{1}{L} \frac{1}{s(s + R/L)} \frac{1}{e^s - 1} \\ &= \frac{1}{R} \left(\frac{1}{s^2} - \frac{L}{R} \frac{1}{s} + \frac{L}{R} \frac{1}{s + R/L} \right) - \frac{1}{R} \left(\frac{1}{s} - \frac{1}{s + R/L} \right) (e^{-s} + e^{-2s} + e^{-3s} + \dots). \end{aligned}$$

Therefore

$$\begin{aligned} i(t) &= \frac{1}{R} \left(t - \frac{L}{R} + \frac{L}{R} e^{-Rt/L} \right) - \frac{1}{R} \left(1 - e^{-R(t-1)/L} \right) \mathcal{U}(t-1) \\ &\quad - \frac{1}{R} \left(1 - e^{-R(t-2)/L} \right) \mathcal{U}(t-2) - \frac{1}{R} \left(1 - e^{-R(t-3)/L} \right) \mathcal{U}(t-3) - \dots \\ &= \frac{1}{R} \left(t - \frac{L}{R} + \frac{L}{R} e^{-Rt/L} \right) - \frac{1}{R} \sum_{n=1}^{\infty} (1 - e^{-R(t-n)/L}) \mathcal{U}(t-n). \end{aligned}$$

The graph of $i(t)$ with $L = 1$ and $R = 1$ is shown below.



57. The differential equation is $x'' + 2x' + 10x = 20f(t)$, where $f(t)$ is the meander function in Problem 49 with

4.4 Additional Operational Properties

$a = \pi$. Using the initial conditions $x(0) = x'(0) = 0$ and taking the Laplace transform we obtain

$$\begin{aligned}(s^2 + 2s + 10) \mathcal{L}\{x(t)\} &= \frac{20}{s}(1 - e^{-\pi s}) \frac{1}{1 + e^{-\pi s}} \\&= \frac{20}{s}(1 - e^{-\pi s})(1 - e^{-\pi s} + e^{-2\pi s} - e^{-3\pi s} + \dots) \\&= \frac{20}{s}(1 - 2e^{-\pi s} + 2e^{-2\pi s} - 2e^{-3\pi s} + \dots) \\&= \frac{20}{s} + \frac{40}{s} \sum_{n=1}^{\infty} (-1)^n e^{-n\pi s}.\end{aligned}$$

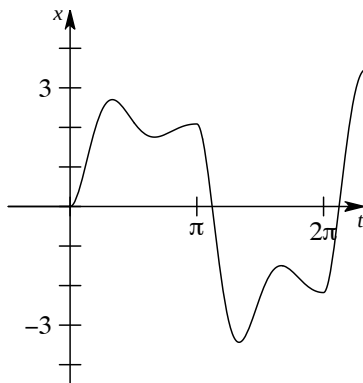
Then

$$\begin{aligned}\mathcal{L}\{x(t)\} &= \frac{20}{s(s^2 + 2s + 10)} + \frac{40}{s(s^2 + 2s + 10)} \sum_{n=1}^{\infty} (-1)^n e^{-n\pi s} \\&= \frac{2}{s} - \frac{2s + 4}{s^2 + 2s + 10} + \sum_{n=1}^{\infty} (-1)^n \left[\frac{4}{s} - \frac{4s + 8}{s^2 + 2s + 10} \right] e^{-n\pi s} \\&= \frac{2}{s} - \frac{2(s + 1) + 2}{(s + 1)^2 + 9} + 4 \sum_{n=1}^{\infty} (-1)^n \left[\frac{1}{s} - \frac{(s + 1) + 1}{(s + 1)^2 + 9} \right] e^{-n\pi s}\end{aligned}$$

and

$$\begin{aligned}x(t) &= 2 \left(1 - e^{-t} \cos 3t - \frac{1}{3} e^{-t} \sin 3t \right) + 4 \sum_{n=1}^{\infty} (-1)^n \left[1 - e^{-(t-n\pi)} \cos 3(t - n\pi) \right. \\&\quad \left. - \frac{1}{3} e^{-(t-n\pi)} \sin 3(t - n\pi) \right] \mathcal{U}(t - n\pi).\end{aligned}$$

The graph of $x(t)$ on the interval $[0, 2\pi)$ is shown below.



58. The differential equation is $x'' + 2x' + x = 5f(t)$, where $f(t)$ is the square wave function with $a = \pi$. Using the initial conditions $x(0) = x'(0) = 0$ and taking the Laplace transform, we obtain

$$\begin{aligned}(s^2 + 2s + 1) \mathcal{L}\{x(t)\} &= \frac{5}{s} \frac{1}{1 + e^{-\pi s}} = \frac{5}{s}(1 - e^{-\pi s} + e^{-2\pi s} - e^{-3\pi s} + e^{-4\pi s} - \dots) \\&= \frac{5}{s} \sum_{n=0}^{\infty} (-1)^n e^{-n\pi s}.\end{aligned}$$

Then

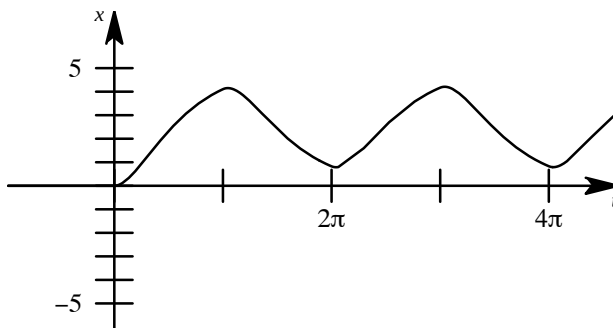
$$\mathcal{L}\{x(t)\} = \frac{5}{s(s + 1)^2} \sum_{n=0}^{\infty} (-1)^n e^{-n\pi s} = 5 \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{s} - \frac{1}{s + 1} - \frac{1}{(s + 1)^2} \right) e^{-n\pi s}$$

4.4 Additional Operational Properties

and

$$x(t) = 5 \sum_{n=0}^{\infty} (-1)^n (1 - e^{-(t-n\pi)} - (t-n\pi)e^{-(t-n\pi)}) \mathcal{U}(t-n\pi).$$

The graph of $x(t)$ on the interval $[0, 4\pi)$ is shown below.



$$59. f(t) = -\frac{1}{t} \mathcal{L}^{-1} \left\{ \frac{d}{ds} [\ln(s-3) - \ln(s+1)] \right\} = -\frac{1}{t} \mathcal{L}^{-1} \left\{ \frac{1}{s-3} - \frac{1}{s+1} \right\} = -\frac{1}{t} (e^{3t} - e^{-t})$$

60. The transform of Bessel's equation is

$$-\frac{d}{ds} [s^2 Y(s) - sy(0) - y'(0)] + sY(s) - y(0) - \frac{d}{ds} Y(s) = 0$$

or, after simplifying and using the initial condition, $(s^2 + 1)Y' + sY = 0$. This equation is both separable and linear. Solving gives $Y(s) = c/\sqrt{s^2 + 1}$. Now $Y(s) = \mathcal{L}\{J_0(t)\}$, where J_0 has a derivative that is continuous and of exponential order, implies by Problem 46 of Exercises 4.2 that

$$1 = J_0(0) = \lim_{s \rightarrow \infty} sY(s) = c \lim_{s \rightarrow \infty} \frac{s}{\sqrt{s^2 + 1}} = c$$

so $c = 1$ and

$$Y(s) = \frac{1}{\sqrt{s^2 + 1}} \quad \text{or} \quad \mathcal{L}\{J_0(t)\} = \frac{1}{\sqrt{s^2 + 1}}.$$

61. (a) Using Theorem 4.8, the Laplace transform of the differential equation is

$$\begin{aligned} & -\frac{d}{ds} [s^2 Y - sy(0) - y'(0)] + sY - y(0) + \frac{d}{ds} [sY - y(0)] + nY \\ &= -\frac{d}{ds} [s^2 Y] + sY + \frac{d}{ds} [sY] + nY \\ &= -s^2 \left(\frac{dY}{ds} \right) - 2sY + sY + s \left(\frac{dY}{ds} \right) + Y + nY \\ &= (s - s^2) \left(\frac{dY}{ds} \right) + (1 + n - s)Y = 0. \end{aligned}$$

Separating variables, we find

$$\frac{dY}{Y} = \frac{1 + n - s}{s^2 - s} ds = \left(\frac{n}{s-1} - \frac{1+n}{s} \right) ds$$

$$\ln Y = n \ln(s-1) - (1+n) \ln s + c$$

$$Y = c_1 \frac{(s-1)^n}{s^{1+n}}.$$

4.4 Additional Operational Properties

Since the differential equation is homogeneous, any constant multiple of a solution will still be a solution, so for convenience we take $c_1 = 1$. The following polynomials are solutions of Laguerre's differential equation:

$$\begin{aligned}
 n = 0: \quad L_0(t) &= \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} = 1 \\
 n = 1: \quad L_1(t) &= \mathcal{L}^{-1} \left\{ \frac{s-1}{s^2} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{1}{s^2} \right\} = 1 - t \\
 n = 2: \quad L_2(t) &= \mathcal{L}^{-1} \left\{ \frac{(s-1)^2}{s^3} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{2}{s^2} + \frac{1}{s^3} \right\} = 1 - 2t + \frac{1}{2}t^2 \\
 n = 3: \quad L_3(t) &= \mathcal{L}^{-1} \left\{ \frac{(s-1)^3}{s^4} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{3}{s^2} + \frac{3}{s^3} - \frac{1}{s^4} \right\} = 1 - 3t + \frac{3}{2}t^2 - \frac{1}{6}t^3 \\
 n = 4: \quad L_4(t) &= \mathcal{L}^{-1} \left\{ \frac{(s-1)^4}{s^5} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{4}{s^2} + \frac{6}{s^3} - \frac{4}{s^4} + \frac{1}{s^5} \right\} \\
 &= 1 - 4t + 3t^2 - \frac{2}{3}t^3 + \frac{1}{24}t^4.
 \end{aligned}$$

(b) Letting $f(t) = t^n e^{-t}$ we note that $f^{(k)}(0) = 0$ for $k = 0, 1, 2, \dots, n-1$ and $f^{(n)}(0) = n!$. Now, by the first translation theorem,

$$\begin{aligned}
 \mathcal{L} \left\{ \frac{e^t}{n!} \frac{d^n}{dt^n} t^n e^{-t} \right\} &= \frac{1}{n!} \mathcal{L} \{ e^t f^{(n)}(t) \} = \frac{1}{n!} \mathcal{L} \{ f^{(n)}(t) \} \Big|_{s \rightarrow s-1} \\
 &= \frac{1}{n!} \left[s^n \mathcal{L} \{ t^n e^{-t} \} - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0) \right]_{s \rightarrow s-1} \\
 &= \frac{1}{n!} \left[s^n \mathcal{L} \{ t^n e^{-t} \} \right]_{s \rightarrow s-1} \\
 &= \frac{1}{n!} \left[s^n \frac{n!}{(s+1)^{n+1}} \right]_{s \rightarrow s-1} = \frac{(s-1)^n}{s^{n+1}} = Y,
 \end{aligned}$$

where $Y = \mathcal{L} \{ L_n(t) \}$. Thus

$$L_n(t) = \frac{e^t}{n!} \frac{d^n}{dt^n} (t^n e^{-t}), \quad n = 0, 1, 2, \dots$$

62. The output for the first three lines of the program are

$$\begin{aligned}
 9y[t] + 6y'[t] + y''[t] &= t \sin[t] \\
 1 - 2s + 9Y + s^2 Y + 6(-2 + sY) &= \frac{2s}{(1 + s^2)^2} \\
 Y &\rightarrow - \left(\frac{-11 - 4s - 22s^2 - 4s^3 - 11s^4 - 2s^5}{(1 + s^2)^2(9 + 6s + s^2)} \right)
 \end{aligned}$$

The fourth line is the same as the third line with $Y \rightarrow$ removed. The final line of output shows a solution involving complex coefficients of e^{it} and e^{-it} . To get the solution in more standard form write the last line as two lines:

$$\begin{aligned}
 \text{euler} &= \{ \mathbf{E}^{\wedge}(\mathbf{It}) \rightarrow \mathbf{Cos}[t] + \mathbf{I Sin}[t], \mathbf{E}^{\wedge}(-\mathbf{It}) \rightarrow \mathbf{Cos}[t] - \mathbf{I Sin}[t] \} \\
 &\text{InverseLaplaceTransform}[Y, s, t] /. \text{euler} // \text{Expand}
 \end{aligned}$$

We see that the solution is

$$y(t) = \left(\frac{487}{250} + \frac{247}{50}t \right) e^{-3t} + \frac{1}{250} (13 \cos t - 15t \cos t - 9 \sin t + 20t \sin t).$$

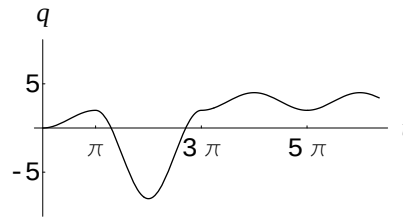
63. The solution is

$$y(t) = \frac{1}{6} e^t - \frac{1}{6} e^{-t/2} \cos \sqrt{15} t - \frac{\sqrt{3/5}}{6} e^{-t/2} \sin \sqrt{15} t.$$

4.4 Additional Operational Properties

64. The solution is

$$q(t) = 1 - \cos t + (6 - 6 \cos t) \mathcal{U}(t - 3\pi) - (4 + 4 \cos t) \mathcal{U}(t - \pi).$$



EXERCISES 4.5

The Dirac Delta Function

1. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s-3} e^{-2s}$$

so that

$$y = e^{3(t-2)} \mathcal{U}(t-2).$$

2. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{2}{s+1} + \frac{e^{-s}}{s+1}$$

so that

$$y = 2e^{-t} + e^{-(t-1)} \mathcal{U}(t-1).$$

3. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s^2+1} (1 + e^{-2\pi s})$$

so that

$$y = \sin t + \sin t \mathcal{U}(t-2\pi).$$

4. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{4} \frac{4}{s^2+16} e^{-2\pi s}$$

so that

$$y = \frac{1}{4} \sin 4(t-2\pi) \mathcal{U}(t-2\pi) = \frac{1}{4} \sin 4t \mathcal{U}(t-2\pi).$$

5. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s^2+1} (e^{-\pi s/2} + e^{-3\pi s/2})$$

so that

$$\begin{aligned} y &= \sin\left(t - \frac{\pi}{2}\right) \mathcal{U}\left(t - \frac{\pi}{2}\right) + \sin\left(t - \frac{3\pi}{2}\right) \mathcal{U}\left(t - \frac{3\pi}{2}\right) \\ &= -\cos t \mathcal{U}\left(t - \frac{\pi}{2}\right) + \cos t \mathcal{U}\left(t - \frac{3\pi}{2}\right). \end{aligned}$$

6. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{s}{s^2 + 1} + \frac{1}{s^2 + 1}(e^{-2\pi s} + e^{-4\pi s})$$

so that

$$y = \cos t + \sin t[\mathcal{U}(t - 2\pi) + \mathcal{U}(t - 4\pi)].$$

7. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s^2 + 2s}(1 + e^{-s}) = \left[\frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{1}{s + 2}\right](1 + e^{-s})$$

so that

$$y = \frac{1}{2} - \frac{1}{2}e^{-2t} + \left[\frac{1}{2} - \frac{1}{2}e^{-2(t-1)}\right]\mathcal{U}(t - 1).$$

8. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{s+1}{s^2(s-2)} + \frac{1}{s(s-2)}e^{-2s} = \frac{3}{4} \frac{1}{s-2} - \frac{3}{4} \frac{1}{s} - \frac{1}{2} \frac{1}{s^2} + \left[\frac{1}{2} \frac{1}{s-2} - \frac{1}{2} \frac{1}{s}\right]e^{-2s}$$

so that

$$y = \frac{3}{4}e^{2t} - \frac{3}{4} - \frac{1}{2}t + \left[\frac{1}{2}e^{2(t-2)} - \frac{1}{2}\right]\mathcal{U}(t - 2).$$

9. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{(s+2)^2 + 1}e^{-2\pi s}$$

so that

$$y = e^{-2(t-2\pi)} \sin t \mathcal{U}(t - 2\pi).$$

10. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{(s+1)^2}e^{-s}$$

so that

$$y = (t-1)e^{-(t-1)}\mathcal{U}(t-1).$$

11. The Laplace transform of the differential equation yields

$$\begin{aligned} \mathcal{L}\{y\} &= \frac{4+s}{s^2+4s+13} + \frac{e^{-\pi s} + e^{-3\pi s}}{s^2+4s+13} \\ &= \frac{2}{3} \frac{3}{(s+2)^2+3^2} + \frac{s+2}{(s+2)^2+3^2} + \frac{1}{3} \frac{3}{(s+2)^2+3^2} (e^{-\pi s} + e^{-3\pi s}) \end{aligned}$$

so that

$$\begin{aligned} y &= \frac{2}{3}e^{-2t} \sin 3t + e^{-2t} \cos 3t + \frac{1}{3}e^{-2(t-\pi)} \sin 3(t-\pi) \mathcal{U}(t-\pi) \\ &\quad + \frac{1}{3}e^{-2(t-3\pi)} \sin 3(t-3\pi) \mathcal{U}(t-3\pi). \end{aligned}$$

12. The Laplace transform of the differential equation yields

$$\begin{aligned} \mathcal{L}\{y\} &= \frac{1}{(s-1)^2(s-6)} + \frac{e^{-2s} + e^{-4s}}{(s-1)(s-6)} \\ &= -\frac{1}{25} \frac{1}{s-1} - \frac{1}{5} \frac{1}{(s-1)^2} + \frac{1}{25} \frac{1}{s-6} + \left[-\frac{1}{5} \frac{1}{s-1} + \frac{1}{5} \frac{1}{s-6}\right] (e^{-2s} + e^{-4s}) \end{aligned}$$

4.5 The Dirac Delta Function

so that

$$y = -\frac{1}{25}e^t - \frac{1}{5}te^t + \frac{1}{25}e^{6t} + \left[-\frac{1}{5}e^{t-2} + \frac{1}{5}e^{6(t-2)}\right]\mathcal{U}(t-2) + \left[-\frac{1}{5}e^{t-4} + \frac{1}{5}e^{6(t-4)}\right]\mathcal{U}(t-4).$$

13. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{2} \frac{2}{s^3} y''(0) + \frac{1}{6} \frac{3!}{s^4} y'''(0) + \frac{1}{6} \frac{P_0}{EI} \frac{3!}{s^4} e^{-Ls/2}$$

so that

$$y = \frac{1}{2} y''(0) x^2 + \frac{1}{6} y'''(0) x^3 + \frac{1}{6} \frac{P_0}{EI} \left(x - \frac{L}{2}\right)^3 \mathcal{U}\left(x - \frac{L}{2}\right).$$

Using $y''(L) = 0$ and $y'''(L) = 0$ we obtain

$$\begin{aligned} y &= \frac{1}{4} \frac{P_0 L}{EI} x^2 - \frac{1}{6} \frac{P_0}{EI} x^3 + \frac{1}{6} \frac{P_0}{EI} \left(x - \frac{L}{2}\right)^3 \mathcal{U}\left(x - \frac{L}{2}\right) \\ &= \begin{cases} \frac{P_0}{EI} \left(\frac{L}{4} x^2 - \frac{1}{6} x^3\right), & 0 \leq x < \frac{L}{2} \\ \frac{P_0 L^2}{4EI} \left(\frac{1}{2} x - \frac{L}{12}\right), & \frac{L}{2} \leq x \leq L. \end{cases} \end{aligned}$$

14. From Problem 13 we know that

$$y = \frac{1}{2} y''(0) x^2 + \frac{1}{6} y'''(0) x^3 + \frac{1}{6} \frac{P_0}{EI} \left(x - \frac{L}{2}\right)^3 \mathcal{U}\left(x - \frac{L}{2}\right).$$

Using $y(L) = 0$ and $y'(L) = 0$ we obtain

$$\begin{aligned} y &= \frac{1}{16} \frac{P_0 L}{EI} x^2 - \frac{1}{12} \frac{P_0}{EI} x^3 + \frac{1}{6} \frac{P_0}{EI} \left(x - \frac{L}{2}\right)^3 \mathcal{U}\left(x - \frac{L}{2}\right) \\ &= \begin{cases} \frac{P_0}{EI} \left(\frac{L}{16} x^2 - \frac{1}{12} x^3\right), & 0 \leq x < \frac{L}{2} \\ \frac{P_0}{EI} \left(\frac{L}{16} x^2 - \frac{1}{12} x^3\right) + \frac{1}{6} \frac{P_0}{EI} \left(x - \frac{L}{2}\right)^3, & \frac{L}{2} \leq x \leq L. \end{cases} \end{aligned}$$

15. You should disagree. Although formal manipulations of the Laplace transform lead to $y(t) = \frac{1}{3}e^{-t} \sin 3t$ in both cases, this function does not satisfy the initial condition $y'(0) = 0$ of the second initial-value problem.

EXERCISES 4.6

Systems of Linear Differential Equations

1. Taking the Laplace transform of the system gives

$$\begin{aligned} s \mathcal{L}\{x\} &= -\mathcal{L}\{x\} + \mathcal{L}\{y\} \\ s \mathcal{L}\{y\} - 1 &= 2 \mathcal{L}\{x\} \end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{1}{(s-1)(s+2)} = \frac{1}{3} \frac{1}{s-1} - \frac{1}{3} \frac{1}{s+2}$$

and

$$\mathcal{L}\{y\} = \frac{1}{s} + \frac{2}{s(s-1)(s+2)} = \frac{2}{3} \frac{1}{s-1} + \frac{1}{3} \frac{1}{s+2}.$$

Then

$$x = \frac{1}{3}e^t - \frac{1}{3}e^{-2t} \quad \text{and} \quad y = \frac{2}{3}e^t + \frac{1}{3}e^{-2t}.$$

2. Taking the Laplace transform of the system gives

$$\begin{aligned} s \mathcal{L}\{x\} - 1 &= 2 \mathcal{L}\{y\} + \frac{1}{s-1} \\ s \mathcal{L}\{y\} - 1 &= 8 \mathcal{L}\{x\} - \frac{1}{s^2} \end{aligned}$$

so that

$$\mathcal{L}\{y\} = \frac{s^3 + 7s^2 - s + 1}{s(s-1)(s^2-16)} = \frac{1}{16} \frac{1}{s} - \frac{8}{15} \frac{1}{s-1} + \frac{173}{96} \frac{1}{s-4} - \frac{53}{160} \frac{1}{s+4}$$

and

$$y = \frac{1}{16} - \frac{8}{15}e^t + \frac{173}{96}e^{4t} - \frac{53}{160}e^{-4t}.$$

Then

$$x = \frac{1}{8}y' + \frac{1}{8}t = \frac{1}{8}t - \frac{1}{15}e^t + \frac{173}{192}e^{4t} + \frac{53}{320}e^{-4t}.$$

3. Taking the Laplace transform of the system gives

$$\begin{aligned} s \mathcal{L}\{x\} + 1 &= \mathcal{L}\{x\} - 2 \mathcal{L}\{y\} \\ s \mathcal{L}\{y\} - 2 &= 5 \mathcal{L}\{x\} - \mathcal{L}\{y\} \end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{-s-5}{s^2+9} = -\frac{s}{s^2+9} - \frac{5}{3} \frac{1}{s^2+9}$$

and

$$x = -\cos 3t - \frac{5}{3} \sin 3t.$$

Then

$$y = \frac{1}{2}x - \frac{1}{2}x' = 2 \cos 3t - \frac{7}{3} \sin 3t.$$

4.6 Systems of Linear Differential Equations

4. Taking the Laplace transform of the system gives

$$\begin{aligned}(s+3)\mathcal{L}\{x\} + s\mathcal{L}\{y\} &= \frac{1}{s} \\ (s-1)\mathcal{L}\{x\} + (s-1)\mathcal{L}\{y\} &= \frac{1}{s-1}\end{aligned}$$

so that

$$\mathcal{L}\{y\} = \frac{5s-1}{3s(s-1)^2} = -\frac{1}{3}\frac{1}{s} + \frac{1}{3}\frac{1}{s-1} + \frac{4}{3}\frac{1}{(s-1)^2}$$

and

$$\mathcal{L}\{x\} = \frac{1-2s}{3s(s-1)^2} = \frac{1}{3}\frac{1}{s} - \frac{1}{3}\frac{1}{s-1} - \frac{1}{3}\frac{1}{(s-1)^2}.$$

Then

$$x = \frac{1}{3} - \frac{1}{3}e^t - \frac{1}{3}te^t \quad \text{and} \quad y = -\frac{1}{3} + \frac{1}{3}e^t + \frac{4}{3}te^t.$$

5. Taking the Laplace transform of the system gives

$$\begin{aligned}(2s-2)\mathcal{L}\{x\} + s\mathcal{L}\{y\} &= \frac{1}{s} \\ (s-3)\mathcal{L}\{x\} + (s-3)\mathcal{L}\{y\} &= \frac{2}{s}\end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{-s-3}{s(s-2)(s-3)} = -\frac{1}{2}\frac{1}{s} + \frac{5}{2}\frac{1}{s-2} - \frac{2}{s-3}$$

and

$$\mathcal{L}\{y\} = \frac{3s-1}{s(s-2)(s-3)} = -\frac{1}{6}\frac{1}{s} - \frac{5}{2}\frac{1}{s-2} + \frac{8}{3}\frac{1}{s-3}.$$

Then

$$x = -\frac{1}{2} + \frac{5}{2}e^{2t} - 2e^{3t} \quad \text{and} \quad y = -\frac{1}{6} - \frac{5}{2}e^{2t} + \frac{8}{3}e^{3t}.$$

6. Taking the Laplace transform of the system gives

$$\begin{aligned}(s+1)\mathcal{L}\{x\} - (s-1)\mathcal{L}\{y\} &= -1 \\ s\mathcal{L}\{x\} + (s+2)\mathcal{L}\{y\} &= 1\end{aligned}$$

so that

$$\mathcal{L}\{y\} = \frac{s+1/2}{s^2+s+1} = \frac{s+1/2}{(s+1/2)^2 + (\sqrt{3}/2)^2}$$

and

$$\mathcal{L}\{x\} = \frac{-3/2}{s^2+s+1} = -\sqrt{3}\frac{\sqrt{3}/2}{(s+1/2)^2 + (\sqrt{3}/2)^2}.$$

Then

$$y = e^{-t/2} \cos \frac{\sqrt{3}}{2}t \quad \text{and} \quad x = -\sqrt{3}e^{-t/2} \sin \frac{\sqrt{3}}{2}t.$$

7. Taking the Laplace transform of the system gives

$$\begin{aligned}(s^2+1)\mathcal{L}\{x\} - \mathcal{L}\{y\} &= -2 \\ -\mathcal{L}\{x\} + (s^2+1)\mathcal{L}\{y\} &= 1\end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{-2s^2-1}{s^4+2s^2} = -\frac{1}{2}\frac{1}{s^2} - \frac{3}{2}\frac{1}{s^2+2}$$

and

$$x = -\frac{1}{2}t - \frac{3}{2\sqrt{2}} \sin \sqrt{2}t.$$

Then

$$y = x'' + x = -\frac{1}{2}t + \frac{3}{2\sqrt{2}} \sin \sqrt{2}t.$$

8. Taking the Laplace transform of the system gives

$$\begin{aligned}(s+1)\mathcal{L}\{x\} + \mathcal{L}\{y\} &= 1 \\ 4\mathcal{L}\{x\} - (s+1)\mathcal{L}\{y\} &= 1\end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{s+2}{s^2+2s+5} = \frac{s+1}{(s+1)^2+2^2} + \frac{1}{2} \frac{2}{(s+1)^2+2^2}$$

and

$$\mathcal{L}\{y\} = \frac{-s+3}{s^2+2s+5} = -\frac{s+1}{(s+1)^2+2^2} + 2 \frac{2}{(s+1)^2+2^2}.$$

Then

$$x = e^{-t} \cos 2t + \frac{1}{2}e^{-t} \sin 2t \quad \text{and} \quad y = -e^{-t} \cos 2t + 2e^{-t} \sin 2t.$$

9. Adding the equations and then subtracting them gives

$$\begin{aligned}\frac{d^2x}{dt^2} &= \frac{1}{2}t^2 + 2t \\ \frac{d^2y}{dt^2} &= \frac{1}{2}t^2 - 2t.\end{aligned}$$

Taking the Laplace transform of the system gives

$$\mathcal{L}\{x\} = 8\frac{1}{s} + \frac{1}{24} \frac{4!}{s^5} + \frac{1}{3} \frac{3!}{s^4}$$

and

$$\mathcal{L}\{y\} = \frac{1}{24} \frac{4!}{s^5} - \frac{1}{3} \frac{3!}{s^4}$$

so that

$$x = 8 + \frac{1}{24}t^4 + \frac{1}{3}t^3 \quad \text{and} \quad y = \frac{1}{24}t^4 - \frac{1}{3}t^3.$$

10. Taking the Laplace transform of the system gives

$$\begin{aligned}(s-4)\mathcal{L}\{x\} + s^3\mathcal{L}\{y\} &= \frac{6}{s^2+1} \\ (s+2)\mathcal{L}\{x\} - 2s^3\mathcal{L}\{y\} &= 0\end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{4}{(s-2)(s^2+1)} = \frac{4}{5} \frac{1}{s-2} - \frac{4}{5} \frac{s}{s^2+1} - \frac{8}{5} \frac{1}{s^2+1}$$

and

$$\mathcal{L}\{y\} = \frac{2s+4}{s^3(s-2)(s^2+1)} = \frac{1}{s} - \frac{2}{s^2} - 2\frac{2}{s^3} + \frac{1}{5} \frac{1}{s-2} - \frac{6}{5} \frac{s}{s^2+1} + \frac{8}{5} \frac{1}{s^2+1}.$$

Then

$$x = \frac{4}{5}e^{2t} - \frac{4}{5} \cos t - \frac{8}{5} \sin t$$

and

$$y = 1 - 2t - 2t^2 + \frac{1}{5}e^{2t} - \frac{6}{5} \cos t + \frac{8}{5} \sin t.$$

4.6 Systems of Linear Differential Equations

11. Taking the Laplace transform of the system gives

$$\begin{aligned}s^2 \mathcal{L}\{x\} + 3(s+1) \mathcal{L}\{y\} &= 2 \\ s^2 \mathcal{L}\{x\} + 3 \mathcal{L}\{y\} &= \frac{1}{(s+1)^2}\end{aligned}$$

so that

$$\mathcal{L}\{x\} = -\frac{2s+1}{s^3(s+1)} = \frac{1}{s} + \frac{1}{s^2} + \frac{1}{2} \frac{2}{s^3} - \frac{1}{s+1}.$$

Then

$$x = 1 + t + \frac{1}{2}t^2 - e^{-t}$$

and

$$y = \frac{1}{3}te^{-t} - \frac{1}{3}x'' = \frac{1}{3}te^{-t} + \frac{1}{3}e^{-t} - \frac{1}{3}.$$

12. Taking the Laplace transform of the system gives

$$\begin{aligned}(s-4) \mathcal{L}\{x\} + 2 \mathcal{L}\{y\} &= \frac{2e^{-s}}{s} \\ -3 \mathcal{L}\{x\} + (s+1) \mathcal{L}\{y\} &= \frac{1}{2} + \frac{e^{-s}}{s}\end{aligned}$$

so that

$$\begin{aligned}\mathcal{L}\{x\} &= \frac{-1/2}{(s-1)(s-2)} + e^{-s} \frac{1}{(s-1)(s-2)} \\ &= \frac{1}{2} \frac{1}{s-1} - \frac{1}{2} \frac{1}{s-2} + e^{-s} \left[-\frac{1}{s-1} + \frac{1}{s-2} \right]\end{aligned}$$

and

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{e^{-s}}{s} + \frac{s/4-1}{(s-1)(s-2)} + e^{-s} \frac{-s/2+2}{(s-1)(s-2)} \\ &= \frac{3}{4} \frac{1}{s-1} - \frac{1}{2} \frac{1}{s-2} + e^{-s} \left[\frac{1}{s} - \frac{3}{2} \frac{1}{s-1} + \frac{1}{s-2} \right].\end{aligned}$$

Then

$$x = \frac{1}{2}e^t - \frac{1}{2}e^{2t} + \left[-e^{t-1} + e^{2(t-1)} \right] \mathcal{U}(t-1)$$

and

$$y = \frac{3}{4}e^t - \frac{1}{2}e^{2t} + \left[1 - \frac{3}{2}e^{t-1} + e^{2(t-1)} \right] \mathcal{U}(t-1).$$

13. The system is

$$\begin{aligned}x_1'' &= -3x_1 + 2(x_2 - x_1) \\ x_2'' &= -2(x_2 - x_1) \\ x_1(0) &= 0 \\ x_1'(0) &= 1 \\ x_2(0) &= 1 \\ x_2'(0) &= 0.\end{aligned}$$

Taking the Laplace transform of the system gives

$$\begin{aligned}(s^2+5) \mathcal{L}\{x_1\} - 2 \mathcal{L}\{x_2\} &= 1 \\ -2 \mathcal{L}\{x_1\} + (s^2+2) \mathcal{L}\{x_2\} &= s\end{aligned}$$

so that

$$\mathcal{L}\{x_1\} = \frac{s^2 + 2s + 2}{s^4 + 7s^2 + 6} = \frac{2}{5} \frac{s}{s^2 + 1} + \frac{1}{5} \frac{1}{s^2 + 1} - \frac{2}{5} \frac{s}{s^2 + 6} + \frac{4}{5\sqrt{6}} \frac{\sqrt{6}}{s^2 + 6}$$

and

$$\mathcal{L}\{x_2\} = \frac{s^3 + 5s + 2}{(s^2 + 1)(s^2 + 6)} = \frac{4}{5} \frac{s}{s^2 + 1} + \frac{2}{5} \frac{1}{s^2 + 1} + \frac{1}{5} \frac{s}{s^2 + 6} - \frac{2}{5\sqrt{6}} \frac{\sqrt{6}}{s^2 + 6}.$$

Then

$$x_1 = \frac{2}{5} \cos t + \frac{1}{5} \sin t - \frac{2}{5} \cos \sqrt{6}t + \frac{4}{5\sqrt{6}} \sin \sqrt{6}t$$

and

$$x_2 = \frac{4}{5} \cos t + \frac{2}{5} \sin t + \frac{1}{5} \cos \sqrt{6}t - \frac{2}{5\sqrt{6}} \sin \sqrt{6}t.$$

14. In this system x_1 and x_2 represent displacements of masses m_1 and m_2 from their equilibrium positions. Since the net forces acting on m_1 and m_2 are

$$-k_1x_1 + k_2(x_2 - x_1) \quad \text{and} \quad -k_2(x_2 - x_1) - k_3x_2,$$

respectively, Newton's second law of motion gives

$$m_1x_1'' = -k_1x_1 + k_2(x_2 - x_1)$$

$$m_2x_2'' = -k_2(x_2 - x_1) - k_3x_2.$$

Using $k_1 = k_2 = k_3 = 1$, $m_1 = m_2 = 1$, $x_1(0) = 0$, $x_1'(0) = -1$, $x_2(0) = 0$, and $x_2'(0) = 1$, and taking the Laplace transform of the system, we obtain

$$(2 + s^2) \mathcal{L}\{x_1\} - \mathcal{L}\{x_2\} = -1$$

$$\mathcal{L}\{x_1\} - (2 + s^2) \mathcal{L}\{x_2\} = -1$$

so that

$$\mathcal{L}\{x_1\} = -\frac{1}{s^2 + 3} \quad \text{and} \quad \mathcal{L}\{x_2\} = \frac{1}{s^2 + 3}.$$

Then

$$x_1 = -\frac{1}{\sqrt{3}} \sin \sqrt{3}t \quad \text{and} \quad x_2 = \frac{1}{\sqrt{3}} \sin \sqrt{3}t.$$

15. (a) By Kirchhoff's first law we have $i_1 = i_2 + i_3$. By Kirchhoff's second law, on each loop we have $E(t) = Ri_1 + L_1i_2'$ and $E(t) = Ri_1 + L_2i_3'$ or $L_1i_2' + Ri_2 + Ri_3 = E(t)$ and $L_2i_3' + Ri_2 + Ri_3 = E(t)$.
 (b) Taking the Laplace transform of the system

$$0.01i_2' + 5i_2 + 5i_3 = 100$$

$$0.0125i_3' + 5i_2 + 5i_3 = 100$$

gives

$$(s + 500) \mathcal{L}\{i_2\} + 500 \mathcal{L}\{i_3\} = \frac{10,000}{s}$$

$$400 \mathcal{L}\{i_2\} + (s + 400) \mathcal{L}\{i_3\} = \frac{8,000}{s}$$

so that

$$\mathcal{L}\{i_3\} = \frac{8,000}{s^2 + 900s} = \frac{80}{9} \frac{1}{s} - \frac{80}{9} \frac{1}{s + 900}.$$

Then

$$i_3 = \frac{80}{9} - \frac{80}{9} e^{-900t} \quad \text{and} \quad i_2 = 20 - 0.0025i_3' - i_3 = \frac{100}{9} - \frac{100}{9} e^{-900t}.$$

4.6 Systems of Linear Differential Equations

(c) $i_1 = i_2 + i_3 = 20 - 20e^{-900t}$

16. (a) Taking the Laplace transform of the system

$$\begin{aligned} i_2' + i_3' + 10i_2 &= 120 - 120\mathcal{U}(t-2) \\ -10i_2' + 5i_3' + 5i_3 &= 0 \end{aligned}$$

gives

$$\begin{aligned} (s+10)\mathcal{L}\{i_2\} + s\mathcal{L}\{i_3\} &= \frac{120}{s}(1-e^{-2s}) \\ -10s\mathcal{L}\{i_2\} + 5(s+1)\mathcal{L}\{i_3\} &= 0 \end{aligned}$$

so that

$$\mathcal{L}\{i_2\} = \frac{120(s+1)}{(3s^2+11s+10)s}(1-e^{-2s}) = \left[\frac{48}{s+5/3} - \frac{60}{s+2} + \frac{12}{s} \right] (1-e^{-2s})$$

and

$$\mathcal{L}\{i_3\} = \frac{240}{3s^2+11s+10}(1-e^{-2s}) = \left[\frac{240}{s+5/3} - \frac{240}{s+2} \right] (1-e^{-2s}).$$

Then

$$i_2 = 12 + 48e^{-5t/3} - 60e^{-2t} - \left[12 + 48e^{-5(t-2)/3} - 60e^{-2(t-2)} \right] \mathcal{U}(t-2)$$

and

$$i_3 = 240e^{-5t/3} - 240e^{-2t} - \left[240e^{-5(t-2)/3} - 240e^{-2(t-2)} \right] \mathcal{U}(t-2).$$

(b) $i_1 = i_2 + i_3 = 12 + 288e^{-5t/3} - 300e^{-2t} - \left[12 + 288e^{-5(t-2)/3} - 300e^{-2(t-2)} \right] \mathcal{U}(t-2)$

17. Taking the Laplace transform of the system

$$\begin{aligned} i_2' + 11i_2 + 6i_3 &= 50 \sin t \\ i_3' + 6i_2 + 6i_3 &= 50 \sin t \end{aligned}$$

gives

$$\begin{aligned} (s+11)\mathcal{L}\{i_2\} + 6\mathcal{L}\{i_3\} &= \frac{50}{s^2+1} \\ 6\mathcal{L}\{i_2\} + (s+6)\mathcal{L}\{i_3\} &= \frac{50}{s^2+1} \end{aligned}$$

so that

$$\mathcal{L}\{i_2\} = \frac{50s}{(s+2)(s+15)(s^2+1)} = -\frac{20}{13} \frac{1}{s+2} + \frac{375}{1469} \frac{1}{s+15} + \frac{145}{113} \frac{s}{s^2+1} + \frac{85}{113} \frac{1}{s^2+1}.$$

Then

$$i_2 = -\frac{20}{13}e^{-2t} + \frac{375}{1469}e^{-15t} + \frac{145}{113}\cos t + \frac{85}{113}\sin t$$

and

$$i_3 = \frac{25}{3}\sin t - \frac{1}{6}i_2' - \frac{11}{6}i_2 = \frac{30}{13}e^{-2t} + \frac{250}{1469}e^{-15t} - \frac{280}{113}\cos t + \frac{810}{113}\sin t.$$

18. Taking the Laplace transform of the system

$$\begin{aligned} 0.5i_1' + 50i_2 &= 60 \\ 0.005i_2' + i_2 - i_1 &= 0 \end{aligned}$$

gives

$$\begin{aligned} s\mathcal{L}\{i_1\} + 100\mathcal{L}\{i_2\} &= \frac{120}{s} \\ -200\mathcal{L}\{i_1\} + (s+200)\mathcal{L}\{i_2\} &= 0 \end{aligned}$$

so that

$$\mathcal{L}\{i_2\} = \frac{24,000}{s(s^2 + 200s + 20,000)} = \frac{6}{5} \frac{1}{s} - \frac{6}{5} \frac{s+100}{(s+100)^2 + 100^2} - \frac{6}{5} \frac{100}{(s+100)^2 + 100^2}.$$

Then

$$i_2 = \frac{6}{5} - \frac{6}{5}e^{-100t} \cos 100t - \frac{6}{5}e^{-100t} \sin 100t$$

and

$$i_1 = 0.005i_2' + i_2 = \frac{6}{5} - \frac{6}{5}e^{-100t} \cos 100t.$$

19. Taking the Laplace transform of the system

$$\begin{aligned} 2i_1' + 50i_2 &= 60 \\ 0.005i_2' + i_2 - i_1 &= 0 \end{aligned}$$

gives

$$\begin{aligned} 2s\mathcal{L}\{i_1\} + 50\mathcal{L}\{i_2\} &= \frac{60}{s} \\ -200\mathcal{L}\{i_1\} + (s+200)\mathcal{L}\{i_2\} &= 0 \end{aligned}$$

so that

$$\begin{aligned} \mathcal{L}\{i_2\} &= \frac{6,000}{s(s^2 + 200s + 5,000)} \\ &= \frac{6}{5} \frac{1}{s} - \frac{6}{5} \frac{s+100}{(s+100)^2 - (50\sqrt{2})^2} - \frac{6\sqrt{2}}{5} \frac{50\sqrt{2}}{(s+100)^2 - (50\sqrt{2})^2}. \end{aligned}$$

Then

$$i_2 = \frac{6}{5} - \frac{6}{5}e^{-100t} \cosh 50\sqrt{2}t - \frac{6\sqrt{2}}{5}e^{-100t} \sinh 50\sqrt{2}t$$

and

$$i_1 = 0.005i_2' + i_2 = \frac{6}{5} - \frac{6}{5}e^{-100t} \cosh 50\sqrt{2}t - \frac{9\sqrt{2}}{10}e^{-100t} \sinh 50\sqrt{2}t.$$

20. (a) Using Kirchhoff's first law we write $i_1 = i_2 + i_3$. Since $i_2 = dq/dt$ we have $i_1 - i_3 = dq/dt$. Using Kirchhoff's second law and summing the voltage drops across the shorter loop gives

$$E(t) = iR_1 + \frac{1}{C}q, \tag{1}$$

so that

$$i_1 = \frac{1}{R_1}E(t) - \frac{1}{R_1C}q.$$

Then

$$\frac{dq}{dt} = i_1 - i_3 = \frac{1}{R_1}E(t) - \frac{1}{R_1C}q - i_3$$

and

$$R_1 \frac{dq}{dt} + \frac{1}{C}q + R_1i_3 = E(t).$$

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Summing the voltage drops across the longer loop gives

$$E(t) = i_1 R_1 + L \frac{di_3}{dt} + R_2 i_3.$$

Combining this with (1) we obtain

$$i_1 R_1 + L \frac{di_3}{dt} + R_2 i_3 = i_1 R_1 + \frac{1}{C} q$$

or

$$L \frac{di_3}{dt} + R_2 i_3 - \frac{1}{C} q = 0.$$

- (b) Using $L = R_1 = R_2 = C = 1$, $E(t) = 50e^{-t} \mathcal{U}(t-1) = 50e^{-1} e^{-(t-1)} \mathcal{U}(t-1)$, $q(0) = i_3(0) = 0$, and taking the Laplace transform of the system we obtain

$$(s+1) \mathcal{L}\{q\} + \mathcal{L}\{i_3\} = \frac{50e^{-1}}{s+1} e^{-s}$$

$$(s+1) \mathcal{L}\{i_3\} - \mathcal{L}\{q\} = 0,$$

so that

$$\mathcal{L}\{q\} = \frac{50e^{-1} e^{-s}}{(s+1)^2 + 1}$$

and

$$q(t) = 50e^{-1} e^{-(t-1)} \sin(t-1) \mathcal{U}(t-1) = 50e^{-t} \sin(t-1) \mathcal{U}(t-1).$$

21. (a) Taking the Laplace transform of the system

$$4\theta_1'' + \theta_2'' + 8\theta_1 = 0$$

$$\theta_1'' + \theta_2'' + 2\theta_2 = 0$$

gives

$$4(s^2 + 2) \mathcal{L}\{\theta_1\} + s^2 \mathcal{L}\{\theta_2\} = 3s$$

$$s^2 \mathcal{L}\{\theta_1\} + (s^2 + 2) \mathcal{L}\{\theta_2\} = 0$$

so that

$$(3s^2 + 4)(s^2 + 4) \mathcal{L}\{\theta_2\} = -3s^3$$

or

$$\mathcal{L}\{\theta_2\} = \frac{1}{2} \frac{s}{s^2 + 4/3} - \frac{3}{2} \frac{s}{s^2 + 4}.$$

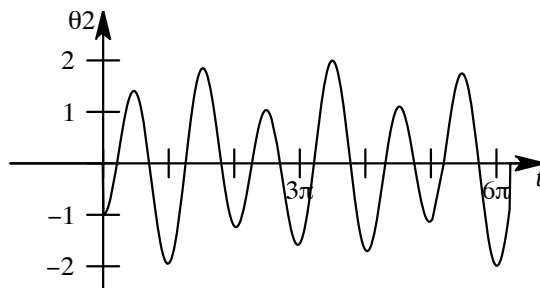
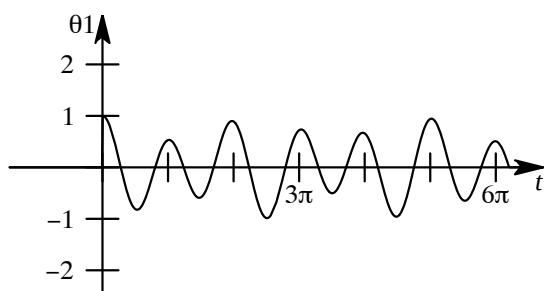
Then

$$\theta_2 = \frac{1}{2} \cos \frac{2}{\sqrt{3}} t - \frac{3}{2} \cos 2t \quad \text{and} \quad \theta_1'' = -\theta_2'' - 2\theta_2$$

so that

$$\theta_1 = \frac{1}{4} \cos \frac{2}{\sqrt{3}} t + \frac{3}{4} \cos 2t.$$

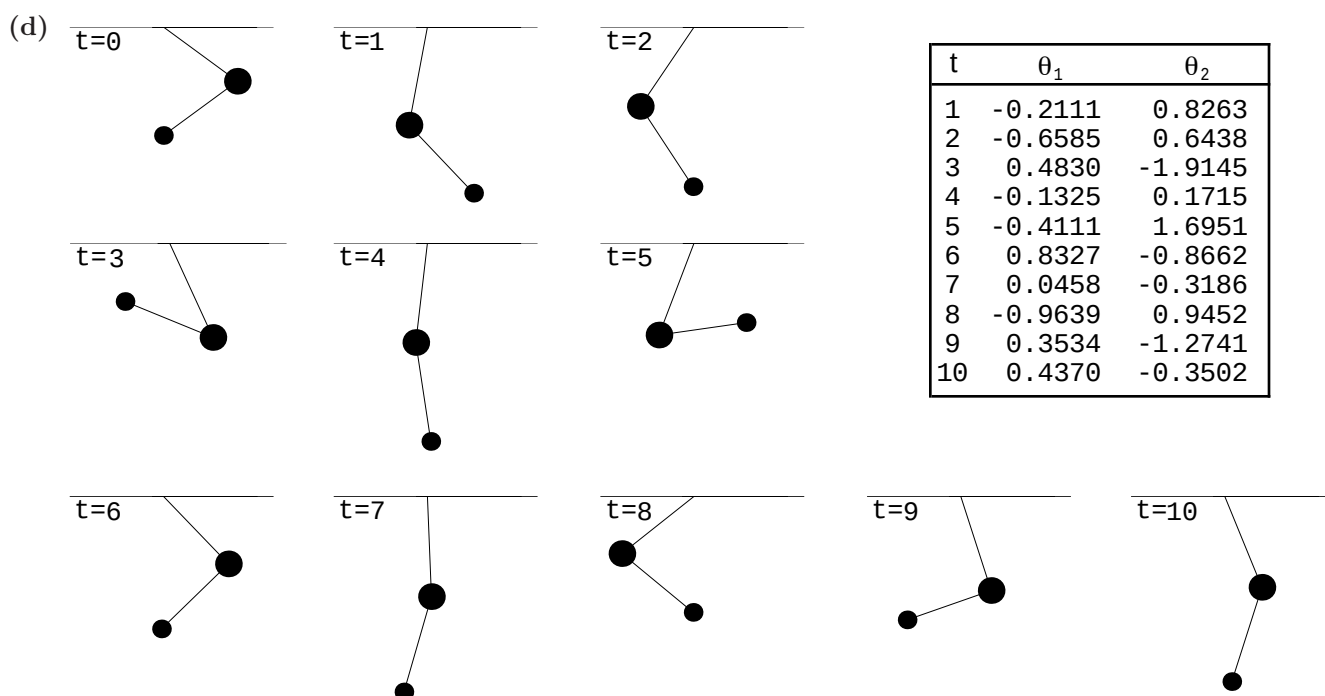
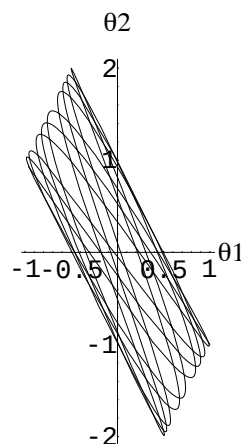
(b)



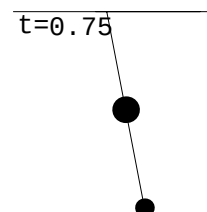
4.6 Systems of Linear Differential Equations

Mass m_2 has extreme displacements of greater magnitude. Mass m_1 first passes through its equilibrium position at about $t = 0.87$, and mass m_2 first passes through its equilibrium position at about $t = 0.66$. The motion of the pendulums is not periodic since $\cos(2t/\sqrt{3})$ has period $\sqrt{3}\pi$, $\cos 2t$ has period π , and the ratio of these periods is $\sqrt{3}$, which is not a rational number.

- (c) The Lissajous curve is plotted for $0 \leq t \leq 30$.



- (e) Using a CAS to solve $\theta_1(t) = \theta_2(t)$ we see that $\theta_1 = \theta_2$ (so that the double pendulum is straight out) when t is about 0.75 seconds.



- (f) To make a movie of the pendulum it is necessary to locate the mass in the plane as a function of time. Suppose that the upper arm is attached to the origin and that the equilibrium position lies along the

4.6 Systems of Linear Differential Equations

negative y -axis. Then mass m_1 is at $(x_1(t), y_1(t))$ and mass m_2 is at $(x_2(t), y_2(t))$, where

$$x_1(t) = 16 \sin \theta_1(t) \quad \text{and} \quad y_1(t) = -16 \cos \theta_1(t)$$

and

$$x_2(t) = x_1(t) + 16 \sin \theta_2(t) \quad \text{and} \quad y_2(t) = y_1(t) - 16 \cos \theta_2(t).$$

A reasonable movie can be constructed by letting t range from 0 to 10 in increments of 0.1 seconds.

CHAPTER 4 REVIEW EXERCISES

1. $\mathcal{L}\{f(t)\} = \int_0^1 t e^{-st} dt + \int_1^\infty (2-t)e^{-st} dt = \frac{1}{s^2} - \frac{2}{s^2} e^{-s}$
2. $\mathcal{L}\{f(t)\} = \int_2^4 e^{-st} dt = \frac{1}{s} (e^{-2s} - e^{-4s})$
3. False; consider $f(t) = t^{-1/2}$.
4. False, since $f(t) = (e^t)^{10} = e^{10t}$.
5. True, since $\lim_{s \rightarrow \infty} F(s) = 1 \neq 0$. (See Theorem 4.5 in the text.)
6. False; consider $f(t) = 1$ and $g(t) = 1$.
7. $\mathcal{L}\{e^{-7t}\} = \frac{1}{s+7}$
8. $\mathcal{L}\{te^{-7t}\} = \frac{1}{(s+7)^2}$
9. $\mathcal{L}\{\sin 2t\} = \frac{2}{s^2+4}$
10. $\mathcal{L}\{e^{-3t} \sin 2t\} = \frac{2}{(s+3)^2+4}$
11. $\mathcal{L}\{t \sin 2t\} = -\frac{d}{ds} \left[\frac{2}{s^2+4} \right] = \frac{4s}{(s^2+4)^2}$
12. $\mathcal{L}\{\sin 2t \mathcal{U}(t-\pi)\} = \mathcal{L}\{\sin 2(t-\pi) \mathcal{U}(t-\pi)\} = \frac{2}{s^2+4} e^{-\pi s}$
13. $\mathcal{L}^{-1}\left\{\frac{20}{s^6}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{6} \frac{5!}{s^6}\right\} = \frac{1}{6} t^5$
14. $\mathcal{L}^{-1}\left\{\frac{1}{3s-1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{3} \frac{1}{s-1/3}\right\} = \frac{1}{3} e^{t/3}$
15. $\mathcal{L}^{-1}\left\{\frac{1}{(s-5)^3}\right\} = \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{2}{(s-5)^3}\right\} = \frac{1}{2} t^2 e^{5t}$
16. $\mathcal{L}^{-1}\left\{\frac{1}{s^2-5}\right\} = \mathcal{L}^{-1}\left\{-\frac{1}{2\sqrt{5}} \frac{1}{s+\sqrt{5}} + \frac{1}{2\sqrt{5}} \frac{1}{s-\sqrt{5}}\right\} = -\frac{1}{2\sqrt{5}} e^{-\sqrt{5}t} + \frac{1}{2\sqrt{5}} e^{\sqrt{5}t}$

17. $\mathcal{L}^{-1}\left\{\frac{s}{s^2-10s+29}\right\} = \mathcal{L}^{-1}\left\{\frac{s-5}{(s-5)^2+2^2} + \frac{5}{2} \frac{2}{(s-5)^2+2^2}\right\} = e^{5t} \cos 2t + \frac{5}{2} e^{5t} \sin 2t$
18. $\mathcal{L}^{-1}\left\{\frac{1}{s^2} e^{-5s}\right\} = (t-5)\mathcal{U}(t-5)$
19. $\mathcal{L}^{-1}\left\{\frac{s+\pi}{s^2+\pi^2} e^{-s}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+\pi^2} e^{-s} + \frac{\pi}{s^2+\pi^2} e^{-s}\right\}$
 $= \cos \pi(t-1)\mathcal{U}(t-1) + \sin \pi(t-1)\mathcal{U}(t-1)$
20. $\mathcal{L}^{-1}\left\{\frac{1}{L^2 s^2 + n^2 \pi^2}\right\} = \frac{1}{L^2} \frac{L}{n\pi} \mathcal{L}^{-1}\left\{\frac{n\pi/L}{s^2 + (n^2 \pi^2)/L^2}\right\} = \frac{1}{Ln\pi} \sin \frac{n\pi}{L} t$
21. $\mathcal{L}\{e^{-5t}\}$ exists for $s > -5$.
22. $\mathcal{L}\{te^{8t}f(t)\} = -\frac{d}{ds}F(s-8)$.
23. $\mathcal{L}\{e^{at}f(t-k)\mathcal{U}(t-k)\} = e^{-ks} \mathcal{L}\{e^{a(t+k)}f(t)\} = e^{-ks}e^{ak} \mathcal{L}\{e^{at}f(t)\} = e^{-k(s-a)}F(s-a)$
24. $\mathcal{L}\left\{\int_0^t e^{a\tau}f(\tau) d\tau\right\} = \frac{1}{s} \mathcal{L}\{e^{at}f(t)\} = \frac{F(s-a)}{s}$, whereas

$$\mathcal{L}\left\{e^{at} \int_0^t f(\tau) d\tau\right\} = \mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} \Big|_{s \rightarrow s-a} = \frac{F(s)}{s} \Big|_{s \rightarrow s-a} = \frac{F(s-a)}{s-a}.$$
25. $f(t)\mathcal{U}(t-t_0)$
26. $f(t) - f(t_0)\mathcal{U}(t-t_0)$
27. $f(t-t_0)\mathcal{U}(t-t_0)$
28. $f(t) - f(t_0)\mathcal{U}(t-t_0) + f(t_0)\mathcal{U}(t-t_0)$
29. $f(t) = t - [(t-1)+1]\mathcal{U}(t-1) + \mathcal{U}(t-1) - \mathcal{U}(t-4) = t - (t-1)\mathcal{U}(t-1) - \mathcal{U}(t-4)$
 $\mathcal{L}\{f(t)\} = \frac{1}{s^2} - \frac{1}{s^2}e^{-s} - \frac{1}{s}e^{-4s}$
 $\mathcal{L}\{e^t f(t)\} = \frac{1}{(s-1)^2} - \frac{1}{(s-1)^2}e^{-(s-1)} - \frac{1}{s-1}e^{-4(s-1)}$
30. $f(t) = \sin t \mathcal{U}(t-\pi) - \sin t \mathcal{U}(t-3\pi) = -\sin(t-\pi)\mathcal{U}(t-\pi) + \sin(t-3\pi)\mathcal{U}(t-3\pi)$
 $\mathcal{L}\{f(t)\} = -\frac{1}{s^2+1}e^{-\pi s} + \frac{1}{s^2+1}e^{-3\pi s}$
 $\mathcal{L}\{e^t f(t)\} = -\frac{1}{(s-1)^2+1}e^{-\pi(s-1)} + \frac{1}{(s-1)^2+1}e^{-3\pi(s-1)}$
31. $f(t) = 2 - 2\mathcal{U}(t-2) + [(t-2)+2]\mathcal{U}(t-2) = 2 + (t-2)\mathcal{U}(t-2)$
 $\mathcal{L}\{f(t)\} = \frac{2}{s} + \frac{1}{s^2}e^{-2s}$
 $\mathcal{L}\{e^t f(t)\} = \frac{2}{s-1} + \frac{1}{(s-1)^2}e^{-2(s-1)}$
32. $f(t) = t - t\mathcal{U}(t-1) + (2-t)\mathcal{U}(t-1) - (2-t)\mathcal{U}(t-2) = t - 2(t-1)\mathcal{U}(t-1) + (t-2)\mathcal{U}(t-2)$
 $\mathcal{L}\{f(t)\} = \frac{1}{s^2} - \frac{2}{s^2}e^{-s} + \frac{1}{s^2}e^{-2s}$
 $\mathcal{L}\{e^t f(t)\} = \frac{1}{(s-1)^2} - \frac{2}{(s-1)^2}e^{-(s-1)} + \frac{1}{(s-1)^2}e^{-2(s-1)}$

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33. Taking the Laplace transform of the differential equation we obtain

$$\mathcal{L}\{y\} = \frac{5}{(s-1)^2} + \frac{1}{2} \frac{2}{(s-1)^3}$$

so that

$$y = 5te^t + \frac{1}{2}t^2e^t.$$

34. Taking the Laplace transform of the differential equation we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{1}{(s-1)^2(s^2-8s+20)} \\ &= \frac{6}{169} \frac{1}{s-1} + \frac{1}{13} \frac{1}{(s-1)^2} - \frac{6}{169} \frac{s-4}{(s-4)^2+2^2} + \frac{5}{338} \frac{2}{(s-4)^2+2^2}\end{aligned}$$

so that

$$y = \frac{6}{169}e^t + \frac{1}{13}te^t - \frac{6}{169}e^{4t}\cos 2t + \frac{5}{338}e^{4t}\sin 2t.$$

35. Taking the Laplace transform of the given differential equation we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{s^3+6s^2+1}{s^2(s+1)(s+5)} - \frac{1}{s^2(s+1)(s+5)}e^{-2s} - \frac{2}{s(s+1)(s+5)}e^{-2s} \\ &= -\frac{6}{25} \cdot \frac{1}{s} + \frac{1}{5} \cdot \frac{1}{s^2} + \frac{3}{2} \cdot \frac{1}{s+1} - \frac{13}{50} \cdot \frac{1}{s+5} \\ &\quad - \left(-\frac{6}{25} \cdot \frac{1}{s} + \frac{1}{5} \cdot \frac{1}{s^2} + \frac{1}{4} \cdot \frac{1}{s+1} - \frac{1}{100} \cdot \frac{1}{s+5} \right) e^{-2s} \\ &\quad - \left(\frac{2}{5} \cdot \frac{1}{s} - \frac{1}{2} \cdot \frac{1}{s+1} + \frac{1}{10} \cdot \frac{1}{s+5} \right) e^{-2s}\end{aligned}$$

so that

$$\begin{aligned}y &= -\frac{6}{25} + \frac{1}{5}t + \frac{3}{2}e^{-t} - \frac{13}{50}e^{-5t} - \frac{4}{25}\mathcal{U}(t-2) - \frac{1}{5}(t-2)\mathcal{U}(t-2) \\ &\quad + \frac{1}{4}e^{-(t-2)}\mathcal{U}(t-2) - \frac{9}{100}e^{-5(t-2)}\mathcal{U}(t-2).\end{aligned}$$

36. Taking the Laplace transform of the differential equation we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{s^3+2}{s^3(s-5)} - \frac{2+2s+s^2}{s^3(s-5)}e^{-s} \\ &= -\frac{2}{125} \frac{1}{s} - \frac{2}{25} \frac{1}{s^2} - \frac{1}{5} \frac{2}{s^3} + \frac{127}{125} \frac{1}{s-5} - \left[-\frac{37}{125} \frac{1}{s} - \frac{12}{25} \frac{1}{s^2} - \frac{1}{5} \frac{2}{s^3} + \frac{37}{125} \frac{1}{s-5} \right] e^{-s}\end{aligned}$$

so that

$$y = -\frac{2}{125} - \frac{2}{25}t - \frac{1}{5}t^2 + \frac{127}{125}e^{5t} - \left[-\frac{37}{125} - \frac{12}{25}(t-1) - \frac{1}{5}(t-1)^2 + \frac{37}{125}e^{5(t-1)} \right] \mathcal{U}(t-1).$$

37. Taking the Laplace transform of the integral equation we obtain

$$\mathcal{L}\{y\} = \frac{1}{s} + \frac{1}{s^2} + \frac{1}{2} \frac{2}{s^3}$$

so that

$$y(t) = 1 + t + \frac{1}{2}t^2.$$

38. Taking the Laplace transform of the integral equation we obtain

$$(\mathcal{L}\{f\})^2 = 6 \cdot \frac{6}{s^4} \quad \text{or} \quad \mathcal{L}\{f\} = \pm 6 \cdot \frac{1}{s^2}$$

so that $f(t) = \pm 6t$.

39. Taking the Laplace transform of the system gives

$$\begin{aligned}s\mathcal{L}\{x\} + \mathcal{L}\{y\} &= \frac{1}{s^2} + 1 \\ 4\mathcal{L}\{x\} + s\mathcal{L}\{y\} &= 2\end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{s^2 - 2s + 1}{s(s-2)(s+2)} = -\frac{1}{4} \frac{1}{s} + \frac{1}{8} \frac{1}{s-2} + \frac{9}{8} \frac{1}{s+2}.$$

Then

$$x = -\frac{1}{4} + \frac{1}{8}e^{2t} + \frac{9}{8}e^{-2t} \quad \text{and} \quad y = -x' + t = \frac{9}{4}e^{-2t} - \frac{1}{4}e^{2t} + t.$$

40. Taking the Laplace transform of the system gives

$$\begin{aligned}s^2\mathcal{L}\{x\} + s^2\mathcal{L}\{y\} &= \frac{1}{s-2} \\ 2s\mathcal{L}\{x\} + s^2\mathcal{L}\{y\} &= -\frac{1}{s-2}\end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{2}{s(s-2)^2} = \frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{1}{s-2} + \frac{1}{(s-2)^2}$$

and

$$\mathcal{L}\{y\} = \frac{-s-2}{s^2(s-2)^2} = -\frac{3}{4} \frac{1}{s} - \frac{1}{2} \frac{1}{s^2} + \frac{3}{4} \frac{1}{s-2} - \frac{1}{(s-2)^2}.$$

Then

$$x = \frac{1}{2} - \frac{1}{2}e^{2t} + te^{2t} \quad \text{and} \quad y = -\frac{3}{4} - \frac{1}{2}t + \frac{3}{4}e^{2t} - te^{2t}.$$

41. The integral equation is

$$10i + 2 \int_0^t i(\tau) d\tau = 2t^2 + 2t.$$

Taking the Laplace transform we obtain

$$\mathcal{L}\{i\} = \left(\frac{4}{s^3} + \frac{2}{s^2}\right) \frac{s}{10s+2} = \frac{s+2}{s^2(5s+2)} = -\frac{9}{s} + \frac{2}{s^2} + \frac{45}{5s+1} = -\frac{9}{s} + \frac{2}{s^2} + \frac{9}{s+1/5}.$$

Thus

$$i(t) = -9 + 2t + 9e^{-t/5}.$$

42. The differential equation is

$$\frac{1}{2} \frac{d^2q}{dt^2} + 10 \frac{dq}{dt} + 100q = 10 - 10\mathcal{U}(t-5).$$

Taking the Laplace transform we obtain

$$\begin{aligned}\mathcal{L}\{q\} &= \frac{20}{s(s^2 + 20s + 200)} (1 - e^{-5s}) \\ &= \left[\frac{1}{10} \frac{1}{s} - \frac{1}{10} \frac{s+10}{(s+10)^2 + 10^2} - \frac{1}{10} \frac{10}{(s+10)^2 + 10^2} \right] (1 - e^{-5s})\end{aligned}$$

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so that

$$q(t) = \frac{1}{10} - \frac{1}{10}e^{-10t} \cos 10t - \frac{1}{10}e^{-10t} \sin 10t \\ - \left[\frac{1}{10} - \frac{1}{10}e^{-10(t-5)} \cos 10(t-5) - \frac{1}{10}e^{-10(t-5)} \sin 10(t-5) \right] \mathcal{U}(t-5).$$

43. Taking the Laplace transform of the given differential equation we obtain

$$\mathcal{L}\{y\} = \frac{2w_0}{EIL} \left(\frac{L}{48} \cdot \frac{4!}{s^5} - \frac{1}{120} \cdot \frac{5!}{s^6} + \frac{1}{120} \cdot \frac{5!}{s^6} e^{-sL/2} \right) + \frac{c_1}{2} \cdot \frac{2!}{s^3} + \frac{c_2}{6} \cdot \frac{3!}{s^4}$$

so that

$$y = \frac{2w_0}{EIL} \left[\frac{L}{48}x^4 - \frac{1}{120}x^5 + \frac{1}{120} \left(x - \frac{L}{2} \right)^5 \mathcal{U} \left(x - \frac{L}{2} \right) + \frac{c_1}{2}x^2 + \frac{c_2}{6}x^3 \right]$$

where $y''(0) = c_1$ and $y'''(0) = c_2$. Using $y''(L) = 0$ and $y'''(L) = 0$ we find

$$c_1 = w_0L^2/24EI, \quad c_2 = -w_0L/4EI.$$

Hence

$$y = \frac{w_0}{12EIL} \left[-\frac{1}{5}x^5 + \frac{L}{2}x^4 - \frac{L^2}{2}x^3 + \frac{L^3}{4}x^2 + \frac{1}{5} \left(x - \frac{L}{2} \right)^5 \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

44. In this case the boundary conditions are $y(0) = y'(0) = 0$ and $y(\pi) = y'(\pi) = 0$. If we let $c_1 = y''(0)$ and $c_2 = y'''(0)$ then

$$s^4 \mathcal{L}\{y\} - s^3y(0) - s^2y'(0) - sy(0) - y'''(0) + 4 \mathcal{L}\{y\} = \mathcal{L}\{\delta(t - \pi/2)\}$$

and

$$\mathcal{L}\{y\} = \frac{c_1}{2} \cdot \frac{2s}{s^4 + 4} + \frac{c_2}{4} \cdot \frac{4}{s^4 + 4} + \frac{w_0}{4EI} \cdot \frac{4}{s^4 + 4} e^{-s\pi/2}.$$

From the table of transforms we get

$$y = \frac{c_1}{2} \sin x \sinh x + \frac{c_2}{4} (\sin x \cosh x - \cos x \sinh x) \\ + \frac{w_0}{4EI} \left[\sin \left(x - \frac{\pi}{2} \right) \cosh \left(x - \frac{\pi}{2} \right) - \cos \left(x - \frac{\pi}{2} \right) \sinh \left(x - \frac{\pi}{2} \right) \right] \mathcal{U} \left(x - \frac{\pi}{2} \right)$$

Using $y(\pi) = 0$ and $y'(\pi) = 0$ we find

$$c_1 = \frac{w_0}{EI} \frac{\sinh \frac{\pi}{2}}{\sinh \pi}, \quad c_2 = -\frac{w_0}{EI} \frac{\cosh \frac{\pi}{2}}{\sinh \pi}.$$

Hence

$$y = \frac{w_0}{2EI} \frac{\sinh \frac{\pi}{2}}{\sinh \pi} \sin x \sinh x - \frac{w_0}{4EI} \frac{\cosh \frac{\pi}{2}}{\sinh \pi} (\sin x \cosh x - \cos x \sinh x) \\ + \frac{w_0}{4EI} \left[\sin \left(x - \frac{\pi}{2} \right) \cosh \left(x - \frac{\pi}{2} \right) - \cos \left(x - \frac{\pi}{2} \right) \sinh \left(x - \frac{\pi}{2} \right) \right] \mathcal{U} \left(x - \frac{\pi}{2} \right).$$

45. (a) With $\omega^2 = g/l$ and $K = k/m$ the system of differential equations is

$$\theta_1'' + \omega^2 \theta_1 = -K(\theta_1 - \theta_2) \\ \theta_2'' + \omega^2 \theta_2 = K(\theta_1 - \theta_2).$$

Denoting the Laplace transform of $\theta(t)$ by $\Theta(s)$ we have that the Laplace transform of the system is

$$(s^2 + \omega^2)\Theta_1(s) = -K\Theta_1(s) + K\Theta_2(s) + s\theta_0 \\ (s^2 + \omega^2)\Theta_2(s) = K\Theta_1(s) - K\Theta_2(s) + s\psi_0.$$

If we add the two equations, we get

$$\Theta_1(s) + \Theta_2(s) = (\theta_0 + \psi_0) \frac{s}{s^2 + \omega^2}$$

which implies

$$\theta_1(t) + \theta_2(t) = (\theta_0 + \psi_0) \cos \omega t.$$

This enables us to solve for first, say, $\theta_1(t)$ and then find $\theta_2(t)$ from

$$\theta_2(t) = -\theta_1(t) + (\theta_0 + \psi_0) \cos \omega t.$$

Now solving

$$\begin{aligned} (s^2 + \omega^2 + K)\Theta_1(s) - K\Theta_2(s) &= s\theta_0 \\ -K\Theta_1(s) + (s^2 + \omega^2 + K)\Theta_2(s) &= s\psi_0 \end{aligned}$$

gives

$$[(s^2 + \omega^2 + K)^2 - K^2]\Theta_1(s) = s(s^2 + \omega^2 + K)\theta_0 + Ks\psi_0.$$

Factoring the difference of two squares and using partial fractions we get

$$\Theta_1(s) = \frac{s(s^2 + \omega^2 + K)\theta_0 + Ks\psi_0}{(s^2 + \omega^2)(s^2 + \omega^2 + 2K)} = \frac{\theta_0 + \psi_0}{2} \frac{s}{s^2 + \omega^2} + \frac{\theta_0 - \psi_0}{2} \frac{s}{s^2 + \omega^2 + 2K},$$

so

$$\theta_1(t) = \frac{\theta_0 + \psi_0}{2} \cos \omega t + \frac{\theta_0 - \psi_0}{2} \cos \sqrt{\omega^2 + 2K} t.$$

Then from $\theta_2(t) = -\theta_1(t) + (\theta_0 + \psi_0) \cos \omega t$ we get

$$\theta_2(t) = \frac{\theta_0 + \psi_0}{2} \cos \omega t - \frac{\theta_0 - \psi_0}{2} \cos \sqrt{\omega^2 + 2K} t.$$

(b) With the initial conditions $\theta_1(0) = \theta_0$, $\theta_1'(0) = 0$, $\theta_2(0) = \theta_0$, $\theta_2'(0) = 0$ we have

$$\theta_1(t) = \theta_0 \cos \omega t, \quad \theta_2(t) = \theta_0 \cos \omega t.$$

Physically this means that both pendulums swing in the same direction as if they were free since the spring exerts no influence on the motion ($\theta_1(t)$ and $\theta_2(t)$ are free of K).

With the initial conditions $\theta_1(0) = \theta_0$, $\theta_1'(0) = 0$, $\theta_2(0) = -\theta_0$, $\theta_2'(0) = 0$ we have

$$\theta_1(t) = \theta_0 \cos \sqrt{\omega^2 + 2K} t, \quad \theta_2(t) = -\theta_0 \cos \sqrt{\omega^2 + 2K} t.$$

Physically this means that both pendulums swing in the opposite directions, stretching and compressing the spring. The amplitude of both displacements is $|\theta_0|$. Moreover, $\theta_1(t) = \theta_0$ and $\theta_2(t) = -\theta_0$ at precisely the same times. At these times the spring is stretched to its maximum.